Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #1 *Topic 1*

A company collects data for temperature, humidity, and atmospheric pressure in cities across multiple continents. The average volume of data that the company collects from each site daily is 500 GB. Each site has a high-speed Internet connection.
The company wants to aggregate the data from all these global sites as quickly as possible in a single Amazon S3 bucket. The solution must minimize operational complexity.
Which solution meets these requirements?

A. Turn on S3 Transfer Acceleration on the destination S3 bucket. Use multipart uploads to directly upload site data to the destination S3 bucket. **Most Voted**

B. Upload the data from each site to an S3 bucket in the closest Region. Use S3 Cross-Region Replication to copy objects to the destination S3 bucket. Then remove the data from the origin S3 bucket.

C. Schedule AWS Snowball Edge Storage Optimized device jobs daily to transfer data from each site to the closest Region. Use S3 Cross-Region Replication to copy objects to the destination S3 bucket.

D. Upload the data from each site to an Amazon EC2 instance in the closest Region. Store the data in an Amazon Elastic Block Store (Amazon EBS) volume. At regular intervals, take an EBS snapshot and copy it to the Region that contains the destination S3 bucket. Restore the EBS volume in that Region.

Correct Answer: *A*

Community vote distribution

A (93%) Other (7%)

Comments

PhucVuu Highly Voted 8 months, 3 weeks ago

Selected Answer: A

Keyword:
From GLOBAL sites as quickly as possible in a SINGLE S3 bucket.
Minimize operational complexity

A. is correct because S3 Transfer Acceleration is support for high speed transfer in Edge location and you can upload it immediately. Also with multipart uploads your big file can be uploaded in parallel.
B, C, D. is not minimize operational and fast when compare to answer A

upvoted 20 times

Ruffyit Highly Voted 8 months, 3 weeks ago

General line: Collect huge amount of the files across multiple continents
Conditions: High speed Internet connectivity
Task: aggregate the data from all in a single S3 bucket
Requirements: as quick as possible, minimize operational complexity

Correct answer A: S3 Transfer Acceleration because:
- ideally works with objects for long-distance transfer (uses Edge Locations)
- can speed up content transfers to and from S3 as much as 50-500%
- use cases: mobile & web application uploads and downloads, distributed office transfers, data exchange with trusted partners. Generally for sharing of large data sets between companies, customers can set up special access to their S3 buckets with accelerated uploads to speed data exchanges and the pace of innovation.

B - about disaster recovery
C - about transferring data between your local environment and the AWS Cloud
D - about disaster recovery

upvoted 12 times

pentium75 1 year, 5 months ago

C, we DO have a "local environment" (a "site") that collect data, and THAT data must go to "the AWS cloud". Why not C? The stem says nothing about large objects, which would be a requirement for the "multipart upload" that is mentioned.

upvoted 3 times

BrantD Most Recent 1 week ago

Selected Answer: A

Least operation complexity.

upvoted 1 times

clfigueiredo12 1 month ago

Selected Answer: U

A. está correto porque o S3 Transfer Acceleration é compatível com a transferência de alta velocidade.

upvoted 1 times

Petter21 1 month ago

Selected Answer: A

A is the correct answer top it exam com

upvoted 1 times

yasitir321 1 month ago

Selected Answer: A

Standard catalog items

Record producers: giving alternative ways of adding information, such as Incidents via the service catalog.
Order guides: to group multiple catalog items in one request.

Click Here: https://surli.cc/mqbyld

The overall catalog is made up of a collection of discrete catalog items.

upvoted 1 times

YDUYGU 1 month, 1 week ago

Selected Answer: A

A is the correct answer

upvoted 1 times

YDUYGU 1 month, 1 week ago

Selected Answer: A

a is the correct answer

upvoted 1 times

topitexamcom 1 month, 1 week ago

Selected Answer: A

Selected Answer: A

i will go with A

upvoted 1 times

Emmamechi 1 month, 2 weeks ago

Selected Answer: A

Great contributions by everyone, A

upvoted 1 times

kfjdaslkfjs 2 months ago

Selected Answer: A

Why A is the best solution:
S3 Transfer Acceleration uses Amazon CloudFront's globally distributed edge locations to speed up uploads to an S3 bucket. This significantly improves upload times from geographically dispersed locations.
Multipart Uploads are ideal for large files (like the 500 GB daily per site), allowing you to upload parts of a file in parallel, improving speed and reliability.

This solution:

Minimizes operational complexity (no need to manage replication or EC2 instances).

Provides high performance by leveraging AWS's optimized global infrastructure.

Is a fully managed solution — no need for manual cleanup or orchestration.

upvoted 1 times

Wylla 2 months, 2 weeks ago

Selected Answer: A

the simplest way

upvoted 1 times

Sreejaggu 2 months, 3 weeks ago

Selected Answer: A

Eliminating other options, Option A is feasible and cost efficient.

upvoted 1 times

ted21 2 months, 3 weeks ago

Selected Answer: A

A for sure

upvoted 1 times

namnv080808 3 months ago

Selected Answer: A

chon A

upvoted 1 times

mc0226 3 months, 1 week ago

Selected Answer: A

S3 Acceleration is more proper with less operation in this case

upvoted 1 times

Dantecito 3 months, 1 week ago

Selected Answer: A

Options B and D are technically possible solutions, but.
B. The cost is high, the process is slower, and it is more complex.
C. This option is not feasible. You would need to purchase Snowball devices, wait for them to arrive, upload the 500GB of data, and then ship them back. This process is not practical for daily use. Snowball devices are designed for one-time uploads of terabytes of data, not for routine or smaller-scale transfers.
D. Using EC2 instances, EBS volumes, and cross-Region snapshots is also not ideal. The cost is significantly higher, and the process is overly complex for this scenario

process is overly complex for this scenario.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #2 *Topic 1*

A company needs the ability to analyze the log files of its proprietary application. The logs are stored in JSON format in an Amazon S3 bucket. Queries will be simple and will run on-demand. A solutions architect needs to perform the analysis with minimal changes to the existing architecture.
What should the solutions architect do to meet these requirements with the LEAST amount of operational overhead?

A. Use Amazon Redshift to load all the content into one place and run the SQL queries as needed.

B. Use Amazon CloudWatch Logs to store the logs. Run SQL queries as needed from the Amazon CloudWatch console.

C. Use Amazon Athena directly with Amazon S3 to run the queries as needed. **Most Voted**

D. Use AWS Glue to catalog the logs. Use a transient Apache Spark cluster on Amazon EMR to run the SQL queries as needed.

Correct Answer: *C*

Community vote distribution

C (99%) A

Comments

airraid2010 Highly Voted 2 years, 8 months ago

Answer: C

Amazon Athena is an interactive query service that makes it easy to analyze data directly in Amazon Simple Storage Service (Amazon S3) using standard SQL. With a few actions in the AWS Management Console, you can point Athena at your data stored in Amazon S3 and begin using standard SQL to run ad-hoc queries and get results in seconds.
upvoted 72 times

BoboChow 2 years, 8 months ago

I agree C is the answer
upvoted 2 times

tt79 2 years, 8 months ago

C is right.
upvoted 1 times

PhucVuu Highly Voted 2 years, 2 months ago

Selected Answer: C

Keyword:
- Queries will be simple and will run on-demand.
- Minimal changes to the existing architecture.
A: Incorrect - We have to do 2 step. load all content to Redshift and run SQL query (This is simple query so we can you Athena, for complex query we will apply Redshit)
B: Incorrect - Our query will be run on-demand so we don't need to use CloudWatch Logs to store the logs.
C: Correct - This is simple query we can apply Athena directly on S3
D: Incorrect - This take 2 step: use AWS Glue to catalog the logs and use Spark to run SQL query
upvoted 45 times

clfigueiredo12 Most Recent 1 month ago

Selected Answer: C

C está certo.
upvoted 1 times

K_SAA 1 month ago

Selected Answer: C

Because the query is simples and run on demand so C is the correct answer
B: use aws cloudwatch is not designed for this use-case because cloudwatch do not support SQL queries and the log json is stored in S3 and cloudwatch can not directly query from s3. Cloudwatch only support its own queries syntax not standard SQL
A: use aws RedShift is not considered the right option since you would need to load all log data into Redshift, which is required more worked and high operational overhead, it goes against the requirement of this use-case queries is simple and running on demand
D: use aws glue and Amazon EMR is more complex in setup and high operational overhead
In conclusion, use Amazon athena is the right option, its simple and serverless meaning no infrastructure to manage.
upvoted 1 times

topitexamcom 1 month, 1 week ago

Selected Answer: C

C is the right answer
upvoted 1 times

Bl_12 3 months, 3 weeks ago

Selected Answer: A

Redshift is query
upvoted 1 times

sumanl75 4 months, 1 week ago

Selected Answer: C

C Correct answer. Athena integrates seamlessly with S3 and allows you to run simple SQL queries in no time. When working with Apache Spark or with SQL in S3, using this service is the best option
upvoted 1 times

ANDREWKIM1 5 months, 2 weeks ago

Selected Answer: C

Amazon Athena meets the requirements for on-demand log analysis with the least operational overhead. It integrates seamlessly with data in Amazon S3, uses SQL for querying, and does not require managing servers or clusters.
upvoted 1 times

MGKYAING 5 months, 2 weeks ago

Selected Answer: C

AWS Athena is a serverless, interactive query service that allows you to analyze data directly in Amazon S3 using standard SQL. It is commonly used for querying logs, performing ad hoc analysis, and building dashboards. Key aspects of AWS Athena include:
Key Features:
1)Serverless:
No infrastructure to manage; pay only for the queries you run.
3)Query S3 Data:
Works directly with structured, semi-structured, and unstructured data stored in S3 (e.g., CSV, JSON, ORC, Parquet).
3)Integration:

3)Integration:
Works seamlessly with AWS Glue for data cataloging, enabling you to query datasets more efficiently.
4)SQL Support:
Built on Presto, Athena supports ANSI SQL.
5)Pay-Per-Query:
Pricing is based on the amount of data scanned per query.
upvoted 1 times

vietqtran 6 months, 2 weeks ago

Selected Answer: C

AWS Athena is an interactive data analytics service from Amazon Web Services (AWS) that allows you to run SQL queries directly on data stored in Amazon S3 without having to manage infrastructure. Athena provides a quick and easy way to analyze data in formats such as CSV, JSON, ORC, Parquet, and Avro without having to move data outside of S3.
upvoted 1 times

yyfabc 7 months ago

not related to this question, but practically, using Redshift Spectrum would be an option here?
upvoted 1 times

zoftdev 6 months, 3 weeks ago

spectrum best for complex query that need parallel/multi processing.
upvoted 1 times

whileloops 8 months ago

Selected Answer: C

Amazon Athena helps you analyze unstructured, semi-structured, and structured data stored in Amazon S3. Examples include CSV, JSON, or columnar data formats such as Apache Parquet and Apache ORC. You can use Athena to run ad-hoc queries using ANSI SQL, without the need to aggregate or load the data into Athena. It uses
Amazon QuickSight for data visualization
AWS Glue Data Catalog allows you to create tables and query data in Athena based on a central metadata store available
upvoted 1 times

cookieMr 8 months, 3 weeks ago

Selected Answer: C

To meet the requirements of analyzing log files stored in JSON format in an Amazon S3 bucket with minimal changes to the existing architecture and minimal operational overhead, the most suitable option would be Option C: Use Amazon Athena directly with Amazon S3 to run the queries as needed.

Amazon Athena is a serverless interactive query service that allows you to analyze data directly from Amazon S3 using standard SQL queries. It eliminates the need for infrastructure provisioning or data loading, making it a low-overhead solution.

Overall, Amazon Athena offers a straightforward and efficient solution for analyzing log files stored in JSON format, ensuring minimal operational overhead and compatibility with simple on-demand queries.
upvoted 3 times

Andreshere 8 months, 3 weeks ago

A. Despite this option can be valid, it implies a bit of operational overhead compared with other options. Additionally, there is no need to aggregate that change to the existing architecture because we are already working in S3, and using other storage services incurs unnecessary costs.
B. To collect the logs, we use CloudTrail over CloudWatch. Running SQL queries from the Amazon CloudWatch console is not recommended for this use case, since it is more used for filtering.
C. Correct answer. Athena integrates seamlessly with S3 and allows you to run simple SQL queries in no time. When working with Apache Spark or with SQL in S3, using this service is the best option.
D. This option incurs elevated operational overhead. Glue is not used to catalog the logs. Analyzing logs with Spark on an EMR cluster is very common, but you can do it faster with the Athena service integrated with S3 directly.
upvoted 4 times

PaulGa 10 months, 1 week ago

C. Athena lets you analyse S3 data using standard SQL. No other steps needed
upvoted 1 times

ChymKuBoy 1 year ago

C for sure

C for sure

upvoted 1 times

Ishu_ 1 year ago

Selected Answer: C

Using Amazon Athena with Amazon S3 is direct and efficient way for querying JSON log files with minimum operational overhead.

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #3 *Topic 1*

A company uses AWS Organizations to manage multiple AWS accounts for different departments. The management account has an Amazon S3 bucket that contains project reports. The company wants to limit access to this S3 bucket to only users of accounts within the organization in AWS Organizations.
Which solution meets these requirements with the LEAST amount of operational overhead?

A. Add the aws PrincipalOrgID global condition key with a reference to the organization ID to the S3 bucket policy. **Most Voted**

B. Create an organizational unit (OU) for each department. Add the aws:PrincipalOrgPaths global condition key to the S3 bucket policy.

C. Use AWS CloudTrail to monitor the CreateAccount, InviteAccountToOrganization, LeaveOrganization, and RemoveAccountFromOrganization events. Update the S3 bucket policy accordingly.

D. Tag each user that needs access to the S3 bucket. Add the aws:PrincipalTag global condition key to the S3 bucket policy.

Correct Answer: *A*

Community vote distribution

A (97%) B

Comments

ude Highly Voted 2 years, 8 months ago

Selected Answer: A

aws:PrincipalOrgID Validates if the principal accessing the resource belongs to an account in your organization.

upvoted 80 times

BoboChow 2 years, 8 months ago

the condition key aws:PrincipalOrgID can prevent the members who don't belong to your organization to access the resource

upvoted 22 times

Naneyerocky Highly Voted 8 months, 3 weeks ago

Selected Answer: A

Condition keys: AWS provides condition keys that you can query to provide more granular control over certain actions.
The following condition keys are especially useful with AWS Organizations:

aws:PrincipalOrgID – Simplifies specifying the Principal element in a resource-based policy. This global key provides an alternative to listing all the account IDs for all AWS accounts in an organization. Instead of listing all of the accounts that are members of an organization, you can specify the organization ID in the Condition element.

aws:PrincipalOrgPaths – Use this condition key to match members of a specific organization root, an OU, or its children. The aws:PrincipalOrgPaths condition key returns true when the principal (root user, IAM user, or role) making the request is in the specified organization path. A path is a text representation of the structure of an AWS Organizations entity.

upvoted 26 times

Sleepy_Lazy_Coder 1 year, 10 months ago

are we not choosing ou because the least overhead term was use? option B also seems correct

upvoted 4 times

EMPERBACH 1 year, 2 months ago

As there are many OU, you need more effort to list down OU path. And question mention about least management overhead to allow users in Organization, not single OU.

upvoted 3 times

BlackMamba_4 1 year, 9 months ago

Exactly

upvoted 1 times

K_SAA Most Recent 1 month ago

Selected Answer: A

A: use aws PrincipalOrgID global condition key with a with a reference to the organization ID to the S3 bucket policy is the correct answer because the company wants to limit access to only user within the organization.

B: is also be considered since the company wanted to limit access to s3 bucket within organization not mention to specific Organizational unit. if you or the company wanted to restrict access to S3 bucket by specific OU consider option B
C: use AWS cloudtrail to monitor action of users within organization, which is more complex and high operational overhead
D: tag each user and use aws:PrincipalTag global condition key to the S3 bucket policy. You would have to tag each user manually, which is required more work overhead

upvoted 1 times

ernie1976 1 month, 1 week ago

Selected Answer: A

This solution es the most simple comparing to other alternatives, just modify one parameter. aws:PrincipalOrgID

upvoted 1 times

Wylla 2 months, 2 weeks ago

Selected Answer: A

aws:PrincipalOrgID - global key provides an alternative to listing all the account IDs for all AWS accounts in an organization.

upvoted 1 times

francisizme 3 months, 1 week ago

Selected Answer: A

Not B because: aws:PrincipalOrgPaths require you change the OU path when the user switch to another OU
Not C because: you need to put effort to update policy which introduces operational overhead.
Not D because: require a good tag management across multiple accounts. Like team "HR", team "QA", team "Security" and so forth. Also need to update if the user is moved to another team with different tag.

upvoted 2 times

Mrigraj12 4 months, 2 weeks ago

Selected Answer: A

PrincipalOrgID is used to validate that the iam user accesing data is from the organisation only

upvoted 1 times

MGKYAING 5 months, 2 weeks ago

Selected Answer: A

aws:PrincipalOrgID is the most efficient and straightforward way to restrict access to resources to entities within an AWS Organization.
It reduces the need for constant monitoring, tagging, or OU management, making it the optimal solution for scenarios requiring minimal operational overhead.

upvoted 1 times

psr83 8 months, 3 weeks ago

Selected Answer: A

use a new condition key, aws:PrincipalOrgID, in these policies to require all principals accessing the resource to be from an account (including the master account) in the organization. For example, let's say you have an Amazon S3 bucket policy and you want to restrict access to only principals from AWS accounts inside of your organization. To accomplish this, you can define the aws:PrincipalOrgID condition and set the value to your organization ID in the bucket policy. Your organization ID is what sets the access control on the S3 bucket. Additionally, when you use this condition, policy permissions apply when you add new accounts to this organization without requiring an update to the policy.

upvoted 2 times

Buruguduystunstugudunstuy 8 months, 3 weeks ago

Selected Answer: A

Answered by ChatGPT with an explanation.

The correct solution that meets these requirements with the least amount of operational overhead is Option A: Add the aws PrincipalOrgID global condition key with a reference to the organization ID to the S3 bucket policy.

Option A involves adding the aws:PrincipalOrgID global condition key to the S3 bucket policy, which allows you to specify the organization ID of the accounts that you want to grant access to the bucket. By adding this condition to the policy, you can limit access to the bucket to only users of accounts within the organization.

upvoted 5 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option B involves creating organizational units (OUs) for each department and adding the aws:PrincipalOrgPaths global condition key to the S3 bucket policy. This option would require more operational overhead, as it involves creating and managing OUs for each department.

Option C involves using AWS CloudTrail to monitor certain events and updating the S3 bucket policy accordingly. While this option could potentially work, it would require ongoing monitoring and updates to the policy, which could increase operational overhead.

upvoted 3 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option D involves tagging each user that needs access to the S3 bucket and adding the aws:PrincipalTag global condition key to the S3 bucket policy. This option would require you to tag each user, which could be time-consuming and could increase operational overhead.

Overall, Option A is the most straightforward and least operationally complex solution for limiting access to the S3 bucket to only users of accounts within the organization.

upvoted 1 times

SilentMilli 8 months, 3 weeks ago

Selected Answer: A

This is the least operationally overhead solution because it requires only a single configuration change to the S3 bucket policy, which will allow access to the bucket for all users within the organization. The other options require ongoing management and maintenance. Option B requires the creation and maintenance of organizational units for each department. Option C requires monitoring of specific CloudTrail events and updates to the S3 bucket policy based on those events. Option D requires the creation and maintenance of tags for each user that needs access to the bucket.

upvoted 1 times

linux_admin 8 months, 3 weeks ago

Selected Answer: A

Option A proposes adding the aws PrincipalOrgID global condition key with a reference to the organization ID to the S3 bucket policy. This would limit access to the S3 bucket to only users of accounts within the organization in AWS Organizations, as the aws PrincipalOrgID condition key can check if the request is coming from within the organization.

upvoted 2 times

martin451 8 months, 3 weeks ago

B. Create an organizational unit (OU) for each department. Add the AWS: Principal Org Paths global condition key to the S3 bucket policy. This solution allows for the S3 bucket to only be accessed by users within the organization in AWS Organizations while minimizing operational overhead by organizing users into OUs and using a single global condition key in the bucket policy. Option A, adding the Principal ID global condition key, would require frequent updates to the policy as new users are added or removed from the organization. Option C, using CloudTrail to monitor events, would require manual updating of the policy based on the events. Option D, tagging each user, would also require manual tagging updates and may not be scalable for larger organizations with many users.

upvoted 1 times

PhucVuu 8 months, 3 weeks ago

Selected Answer: A

Keywords:
- Company uses AWS Organizations
- Limit access to this S3 bucket to only users of accounts within the organization in AWS Organizations
- LEAST amount of operational overhead
A: Correct - We just add PrincipalOrgID global condition key with a reference to the organization ID to the S3 bucket policy
B: Incorrect - We can limit access by this way but this will take more amount of operational overhead
C: Incorrect - AWS CloudTrail only log API events, we can not prevent user access to S3 bucket. For update S3 bucket policy to make it work you should manually add each account -> this way will not be cover in case of new user is added to Organization.
D: Incorrect - We can limit access by this way but this will take most amount of operational overhead

upvoted 12 times

cookieMr 8 months, 3 weeks ago

Selected Answer: A

Option A, which suggests adding the aws PrincipalOrgID global condition key with a reference to the organization ID to the S3 bucket policy, is a valid solution to limit access to the S3 bucket to users within the organization in AWS Organizations. It can effectively achieve the desired access control.

It restricts access to the S3 bucket based on the organization ID, ensuring that only users within the organization can access the bucket. This method is suitable if you want to restrict access at the organization level rather than individual departments or organizational units.

The operational overhead for Option A is also relatively low since it involves adding a global condition key to the S3 bucket policy. However, it is important to note that the organization ID must be accurately configured in the bucket policy to ensure the desired access control is enforced.

In summary, Option A is a valid solution with minimal operational overhead that can limit access to the S3 bucket to users within the organization using the aws PrincipalOrgID global condition key.

upvoted 1 times

Ruffyit 8 months, 3 weeks ago

AWS Identity and Access Management (IAM) now makes it easier for you to control access to your AWS resources by using the AWS organization of IAM principals (users and roles). For some services, you grant permissions using resource-based policies to specify the accounts and principals that can access the resource and what actions they can perform on it. Now, you can use a new condition key, aws:PrincipalOrgID, in these policies to require all principals accessing the resource to be from an account (including the master account) in the organization.

upvoted 4 times

Andreshere 8 months, 3 weeks ago

Selected Answer: A

A. Correct answer. Bucket policy controls who can access to S3 and their objects. If we refer in the bucket policy to the organization, we can limit who can access inside that organization.
B. Despite this option is correct, it is unnecessarily complex. We don't need to separate the AWS Organization users for the requirements imposed in the question. So, it only aggregates more operational overhead.
C. Using CloudTrail for controlling the S3 access permissions is not suitable and require so many events to be monitored. Additionally, it only registers the logs, so CloudTrail cannot impose restrictions over the accounts that access to S3.
D. Tagging each user is not an scalable or efficient solution since you need to tag every user in the infrastructure, which is probably not static. Additionally, it makes unnecessary verbose the S3 bucket policy associated to that bucket.

upvoted 7 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #4 *Topic 1*

An application runs on an Amazon EC2 instance in a VPC. The application processes logs that are stored in an Amazon S3 bucket. The EC2 instance needs to access the S3 bucket without connectivity to the internet.
Which solution will provide private network connectivity to Amazon S3?

A. Create a gateway VPC endpoint to the S3 bucket. **Most Voted**

B. Stream the logs to Amazon CloudWatch Logs. Export the logs to the S3 bucket.

C. Create an instance profile on Amazon EC2 to allow S3 access.

D. Create an Amazon API Gateway API with a private link to access the S3 endpoint.

Correct Answer: *A*

Community vote distribution

A (99%) D

Comments

PhucVuu Highly Voted 8 months, 3 weeks ago

Selected Answer: A

Keywords:
- EC2 in VPC
- EC2 instance needs to access the S3 bucket without connectivity to the internet

A: Correct - Gateway VPC endpoint can connect to S3 bucket privately without additional cost
B: Incorrect - You can set up interface VPC endpoint for CloudWatch Logs for private network from EC2 to CloudWatch. But from CloudWatch to S3 bucket: Log data can take up to 12 hours to become available for export and the requirement only need EC2 to S3
C: Incorrect - Create an instance profile just grant access but not help EC2 connect to S3 privately
D: Incorrect - API Gateway like the proxy which receive network from out site and it forward request to AWS Lambda, Amazon EC2, Elastic Load Balancing products such as Application Load Balancers or Classic Load Balancers, Amazon DynamoDB, Amazon Kinesis, or any publicly available HTTPS-based endpoint. But not S3

upvoted 71 times

rxwcl 7 months, 3 weeks ago

Option C involves creating an instance profile on the EC2 instance to allow S3 access. While this option could potentially work, it would not provide private network connectivity to S3, as the EC2 instance would still need to access S3 over the internet.

Option D involves creating an Amazon API Gateway API with a private link to access the S3 endpoint. This option would not provide private network connectivity to S3, as the API Gateway API is not a network interface that can be used to privately connect to S3.

Overall, Option A is the correct solution for providing private network connectivity to Amazon S3 from an EC2 instance in a VPC.

upvoted 2 times

Austinlorenzmccoy 1 year, 6 months ago

Thank you so much

upvoted 1 times

D2w Highly Voted 2 years, 8 months ago

Selected Answer: A

VPC endpoint allows you to connect to AWS services using a private network instead of using the public Internet

upvoted 32 times

K_SAA Most Recent 1 month ago

Selected Answer: A

Keywords:
log store in s3, Ec2 instance need to access to s3 bucket without internet
A: use gateway vpc endpoint allow private network connectivity to s3 bucket without publishing to internet
B: Stream the logs to Amazon CloudWatch Logs. Export the logs to the S3 bucket since the question need ec2 instance stay in vpc access to s3 bucket privately but this way do not grant access to s3 bucket
C: Create an instance profile on Amazon EC2 to allow S3 access. This way only authorization access to s3 bucket, not grant private network connectivity to s3 bucket since this also need internet access
D. Create an Amazon API Gateway API with a private link to access the S3 endpoint. Since API gateway acts like a proxy receive requests from outside and forward to the request to aws services like lambda, etc, not using for s3 access

upvoted 1 times

ernie1976 1 month, 1 week ago

Selected Answer: A

It is a simple communication between EC2 and S3, it is no needed to go out to internet, so and endopint is the best solution.

upvoted 1 times

Palee 1 month, 2 weeks ago

Selected Answer: A

A is correct

upvoted 1 times

melvis8 2 months, 3 weeks ago

Selected Answer: A

The correct answer is A because a gateway endpoint allows us to securely and cost efficiently access either an S3 or Amazon DynamoDB database within our VPC

upvoted 1 times

Bl_12 3 months, 3 weeks ago

Selected Answer: D

API gateway

upvoted 1 times

Aayush_786 4 months, 2 weeks ago

Selected Answer: A

If accessing S3/DynamoDB privately: Use a Gateway VPC Endpoint.
If accessing any other AWS service privately: Use an Interface Endpoint (via PrivateLink).
If accessing third-party SaaS applications or services in another account/VPC: Use PrivateLink.

upvoted 2 times

Chumi 4 months, 2 weeks ago

Selected Answer: A

option A vpc endpoint can easily connect to an S3 bucket privately with little or zero cost accrued.

upvoted 1 times

Mrigraj12 4 months, 2 weeks ago

Selected Answer: A

Create gateway endpoint to access s3 bucket so as the ec2 will not require to go over the internet to access s3 bucket and the process will be fast and cheap also!

upvoted 1 times

MGKYAING 5 months, 2 weeks ago

Selected Answer: A

A Gateway VPC Endpoint is designed to provide private network connectivity between resources in a VPC (such as EC2 instances) and services like Amazon S3 or DynamoDB without requiring an internet gateway, NAT gateway, or NAT instance.
When a gateway VPC endpoint is set up for S3:
Traffic between the EC2 instance and the S3 bucket stays within the AWS private network.
This ensures secure, cost-efficient, and private access to the S3 bucket without requiring public internet connectivity.

upvoted 2 times

cookieMr 8 months, 3 weeks ago

Selected Answer: A

Here's why Option A is the correct choice:

Gateway VPC Endpoint: A gateway VPC endpoint allows you to privately connect your VPC to supported AWS services. By creating a gateway VPC endpoint for S3, you can establish a private connection between your VPC and the S3 service without requiring internet connectivity.

Private network connectivity: The gateway VPC endpoint for S3 enables your EC2 instance within the VPC to access the S3 bucket over the private network, ensuring secure and direct communication between the EC2 instance and S3.

No internet connectivity required: Since the requirement is to access the S3 bucket without internet connectivity, the gateway VPC endpoint provides a private and direct connection to S3 without needing to route traffic through the internet.

Minimal operational complexity: Setting up a gateway VPC endpoint is a straightforward process. It involves creating the endpoint and configuring the appropriate routing in the VPC. This solution minimizes operational complexity while providing the required private network connectivity.

upvoted 4 times

Ruffyit 8 months, 3 weeks ago

Keywords:
- EC2 in VPC
- EC2 instance needs to access the S3 bucket without connectivity to the internet
VPC endpoint allows you to connect to AWS services using a private network instead of using the public Internet.

With a gateway endpoint, you can access Amazon S3 from your VPC, without requiring an internet gateway or NAT device for your VPC, and with no additional cost. However, gateway endpoints do not allow access from on-premises networks, from peered VPCs in other AWS Regions, or through a transit gateway.

upvoted 4 times

Andreshere 8 months, 3 weeks ago

Selected Answer: A

A. Correct answer. The easiest way to get private network connectivity in S3 is using VPC gateway endpoint. This service is free, and it is integrated natively with S3.
B. Amazon CloudWatch Logs only collects and monitors logs but natively has not mechanisms to use private connection.
C. Instance profiles are used to assign IAM roles to an EC2 instance, but it is not related to network connectivity.
D. API Gateway like the proxy which receive network from out site and it forward request to AWS Lambda, Amazon EC2, Elastic Load Balancing products such as Application Load Balancers or Classic Load Balancers, Amazon DynamoDB, Amazon Kinesis, or any publicly available HTTPS-based endpoint. But not S3.

upvoted 6 times

Buruguduystunstugudunstuy 8 months, 3 weeks ago

Selected Answer: A

CORRECT ANSWER
The correct solution that will provide private network connectivity to Amazon S3 is Option A: Create a gateway VPC endpoint to the S3 bucket.

EXPLANATION
Option A involves creating a gateway VPC endpoint, which is a network interface in a VPC that allows you to privately connect to a service over the Amazon network. You can create a gateway VPC endpoint for Amazon S3, which will allow the EC2 instance in the VPC to access the S3 bucket without connectivity to the internet.

Option B involves streaming the logs to Amazon CloudWatch Logs and then exporting the logs to the S3 bucket. This option would not provide private network connectivity to S3, as the logs would need to be exported over the internet.

upvoted 3 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option C involves creating an instance profile on the EC2 instance to allow S3 access. While this option could potentially work, it would not provide private network connectivity to S3, as the EC2 instance would still need to access S3 over the internet.

Option D involves creating an Amazon API Gateway API with a private link to access the S3 endpoint. This option would not provide private network connectivity to S3, as the API Gateway API is not a network interface that can be used to privately connect to S3.

Overall, Option A is the correct solution for providing private network connectivity to Amazon S3 from an EC2 instance in a VPC.

upvoted 1 times

Chiaki35 10 months ago

A. You should create VPC endpoint and link to S3 endpoint to transfer internally in AWS without internet

upvoted 1 times

PaulGa 10 months, 1 week ago

Ans A. VPC = Virtual Private Cloud, so its already private... so just create another end point...

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #5 *Topic 1*

A company is hosting a web application on AWS using a single Amazon EC2 instance that stores user-uploaded documents in an Amazon EBS volume. For better scalability and availability, the company duplicated the architecture and created a second EC2 instance and EBS volume in another Availability Zone, placing both behind an Application Load Balancer. After completing this change, users reported that, each time they refreshed the website, they could see one subset of their documents or the other, but never all of the documents at the same time.
What should a solutions architect propose to ensure users see all of their documents at once?

A. Copy the data so both EBS volumes contain all the documents

B. Configure the Application Load Balancer to direct a user to the server with the documents

C. Copy the data from both EBS volumes to Amazon EFS. Modify the application to save new documents to Amazon EFS
Most Voted

D. Configure the Application Load Balancer to send the request to both servers. Return each document from the correct server

Correct Answer: *C*

Community vote distribution

C (99%) D

Comments

D2w Highly Voted 2 years, 8 months ago
Selected Answer: C

Concurrent or at the same time key word for EFS
upvoted 56 times

mikey2000 Highly Voted 2 years, 7 months ago

Ebs doesnt support cross az only reside in one Az but Efs does, that why it's c
upvoted 41 times

pbpally 2 years, 1 month ago

And just for clarification to others, you can have COPIES of the same EBS volume in one AZ and in another via EBS Snapshots, but don't confuse that with the idea of having some sort of global capability that has concurrent copying mechanisms.

upvoted 10 times

K_SAA Most Recent 1 month ago

Selected Answer: C

Problem
each EC2 instance has its own EBS volume, and EBS volume is specific for AZ specific
so when user send request to specific ec2 instance they only see the response from each EBS volume not all because EBS volume do not sever for cross region problem

C: use EFS because EFS a fully managed and network file system can be mounted to mutiple ec2 instances across mutiple AZ so user can see all documents response from EFS instead of subset of documents

A. Copy the data so both EBS volumes contain all the documents is not correct because when user send request to one instance on a specific AZ they only see the documents from that EBS AZ not both

upvoted 1 times

Palee 1 month, 2 weeks ago

Selected Answer: C

C is correct

upvoted 1 times

melvis8 2 months, 3 weeks ago

Selected Answer: C

The correct anszwer to this question is C question EBS volumes are AZ locked so when transferring the architecture to a different AZ the data can no longer be received by users whereas with EFS volume we can EC2 instances across AZs

upvoted 1 times

kimardamina 2 months, 4 weeks ago

Selected Answer: C

My immediate understanding was the fact that EFS is multi az and will make it less complex as a solution for this.

upvoted 1 times

francisizme 3 months, 1 week ago

Selected Answer: C

Not A: because it requires sync between the EBS volumes which is complex and not scalable.
Not B: It's not scalable if the LB only directs user to one instance
Not D: Impractical. The application will need to have a mechanism to merge the responses coming from 2 instance

upvoted 1 times

Mrigraj12 4 months, 2 weeks ago

Selected Answer: C

Use common storage like EFS so that the instances can access the document without any hassle and without being uploaded into them separately

upvoted 1 times

ANDREWKIM1 5 months, 2 weeks ago

Selected Answer: C

Amazon S3 is the optimal solution for this scenario. It provides centralized, scalable, and highly available storage, ensuring that all users can see all their documents regardless of which instance they are routed to.

upvoted 1 times

Guru4Cloud 8 months, 3 weeks ago

Selected Answer: C

The answer is C. Copy the data from both EBS volumes to Amazon EFS. Modify the application to save new documents to Amazon EFS.

The current architecture is using two separate EBS volumes, one for each EC2 instance. This means that each instance only has a subset of the documents. When a user refreshes the website, the Application Load Balancer will randomly direct them to one of the two instances. If the user's documents are not on the instance that they are directed to, they will not be able to see them.

upvoted 5 times

Buruguduystunstugudunstuy 8 months, 3 weeks ago

Selected Answer: C

To ensure that users see all of their documents at once, the solutions architect should propose Option C: Copy the data from both EBS volumes to Amazon EFS. Modify the application to save new documents to Amazon EFS.

Option C involves copying the data from both EBS volumes to Amazon Elastic File System (EFS), and modifying the application to save new documents to EFS. Amazon EFS is a fully managed, scalable file storage service that allows you to store and access files from multiple EC2 instances concurrently. By moving the data to EFS and modifying the application to save new documents to EFS, the application will be able to access all of the documents from a single, centralized location, ensuring that users see all of their documents at once.

Overall, Option C is the most effective solution for ensuring that users see all of their documents at once.

upvoted 6 times

Buruguduystunstugudunstuy 2 years, 5 months ago

WRONG
Option A involves copying the data so both EBS volumes contain all the documents. This option would not solve the issue, as the data is still stored on two separate EBS volumes, and the application would still need to read from both volumes to retrieve all of the documents.

Option B involves configuring the Application Load Balancer to direct a user to the server with the documents. This option would not solve the issue, as the user may not always be directed to the server that has the documents they are looking for.

Option D involves configuring the Application Load Balancer to send the request to both servers and return each document from the correct server. This option would not be an efficient solution, as it would require the application to send requests to both servers and receive and process the responses from both servers, which could increase the load on the application.

upvoted 4 times

PhucVuu 8 months, 3 weeks ago

Selected Answer: C

Keyword:
second EC2 instance and EBS volume. They could see one subset of their documents or the other, but never all of the documents at the same time.

EBS: attached to one instance (special EBS io1, io2 can attached to multiple instances but not much)
EFS: can attached to multiple instances

A: Incorrect - EBS volumes don't have function to copy data from running EBS volume to running EBS volume.
B: Incorrect - We can use sticky session to forward same user to the same server but when user lose the session the user might be forward to another server.
C: Correct - Because 2 instance now point to one EFS data storage, user will see both data.
D: Incorrect - We only use Traffic Mirroring to sent request to both servers. Application Load Balancer don't support send request to both servers because it's design it balance workload between server. And also ALB cannot combine document from both servers and return.

upvoted 10 times

IMTechguy 8 months, 3 weeks ago

Option A is not a good solution because copying data to both volumes would not ensure consistency of the data.

Option B would require the Load Balancer to have knowledge of which documents are stored on which server, which would be difficult to maintain.

Option C is a viable solution, but may require modifying the application to use Amazon EFS instead of EBS.

Option D is a good solution because it would distribute the requests to both servers and return the correct document from the correct server. This can be achieved by configuring session stickiness on the Load Balancer so that each user's requests are directed to the same server for consistency.

Therefore, the correct answer is D.

upvoted 3 times

Anthony_Rodrigues 8 months ago

Sending requests to both servers can increase the response time since it would require checking two servers instead of one.
Session stickiness only works if the user has data in only one of the servers; otherwise, it would continue missing data.
Option C is not the best option, but is the one that fits better.

upvoted 1 times

PaulGa 10 months, 1 week ago

Ans C. Altho' A could do it, it would require a manual operation; the clue is "better scalability and availability" - EFS does that automatically

upvoted 1 times

aquarian_ngc 11 months, 1 week ago

Correct Answer is C

upvoted 2 times

williamsmith95 11 months, 3 weeks ago

Selected Answer: C

C is correct answer.

upvoted 3 times

ChymKuBoy 1 year ago

Selected Answer: C

C for sure

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #6 *Topic 1*

A company uses NFS to store large video files in on-premises network attached storage. Each video file ranges in size from 1 MB to 500 GB. The total storage is 70 TB and is no longer growing. The company decides to migrate the video files to Amazon S3. The company must migrate the video files as soon as possible while using the least possible network bandwidth.
Which solution will meet these requirements?

A. Create an S3 bucket. Create an IAM role that has permissions to write to the S3 bucket. Use the AWS CLI to copy all files locally to the S3 bucket.

B. Create an AWS Snowball Edge job. Receive a Snowball Edge device on premises. Use the Snowball Edge client to transfer data to the device. Return the device so that AWS can import the data into Amazon S3. Most Voted

C. Deploy an S3 File Gateway on premises. Create a public service endpoint to connect to the S3 File Gateway. Create an S3 bucket. Create a new NFS file share on the S3 File Gateway. Point the new file share to the S3 bucket. Transfer the data from the existing NFS file share to the S3 File Gateway.

D. Set up an AWS Direct Connect connection between the on-premises network and AWS. Deploy an S3 File Gateway on premises. Create a public virtual interface (VIF) to connect to the S3 File Gateway. Create an S3 bucket. Create a new NFS file share on the S3 File Gateway. Point the new file share to the S3 bucket. Transfer the data from the existing NFS file share to the S3 File Gateway.

Correct Answer: *B*

Community vote distribution

B (88%) Other (12%)

Comments

Gatt Highly Voted 8 months, 3 weeks ago

Selected Answer: B

Let's analyse this:

B. On a Snowball Edge device you can copy files with a speed of up to 100Gbps. 70TB will take around 5600 seconds, so very quickly, less than 2 hours. The downside is that it'll take between 4-6 working days to receive the device and then another 2-3 working days to send it back and for AWS to move the data onto S3 once it reaches them. Total time: 6-9 working days. Bandwidth used: 0.

C. File Gateway uses the Internet, so maximum speed will be at most 1Gbps, so it'll take a minimum of 6.5 days and you use 70TB of Internet bandwidth.

D. You can achieve speeds of up to 10Gbps with Direct Connect. Total time 15.5 hours and you will use 70TB of bandwidth. However, what's interesting is that the question does not specific what type of bandwidth? Direct Connect does not use your Internet bandwidth, as you will have a dedicate peer to peer connectivity between your on-prem and the AWS Cloud, so technically, you're not using your "public" bandwidth.

The requirements are a bit too vague but I think that B is the most appropriate answer, although D might also be correct if the bandwidth usage refers strictly to your public connectivity.

upvoted 126 times

abhishek_m89 2 years, 6 months ago

and it says, "The total storage is 70 TB and is no longer growing". Thats why it should be B.

upvoted 7 times

YDUYGU 2 months, 1 week ago

DX Lead times are often longer than I month to establish a new connection.That's why D is the wrong answer on the other hand.

upvoted 3 times

Gatt 2 years, 7 months ago

I will add that the question does not specify if the company already has DA in place or not. If they don't have DA in place, it will take a long time (weeks) for DA connectivity to be setup. Another point for B here, as Snowball is much quicker from this perspective.

upvoted 10 times

pentium75 1 year, 5 months ago

It does, because option D says "SET UP a DirectConnect connection", not "use an existing DirectConnect connection".

upvoted 6 times

OBIOHAnze 1 year ago

B is the correct answer because the migration needs to be completed as soon as possible with limited bandwidth

upvoted 3 times

tuloveu Highly Voted 2 years, 8 months ago

Selected Answer: B

As using the least possible network bandwidth.

upvoted 35 times

K_SAA Most Recent 1 month ago

Selected Answer: B

Keywords: large volume data, least network bandwidth and volume data no longer growing
B: use aws snowball edge job is more suitable way to migrate large amount of data with least network bandwidth and very fast.
You setup aws snowball edge job physically and copy data to aws snowball edge job and send it to aws and then aws will uploads that for you to s3 bucket
C is incorrect since this also offer a high network bandwidth to upload
D is incorrect since this require heavy setup, take a week to set up aws direct connect and still network bandwidth
A: take a longer to upload this large volume data so this option is incorrect

upvoted 1 times

ernie1976 1 month, 1 week ago

Selected Answer: B

It is needed to do with urgency, so no more configuration for using internet or expensive connection as direct connect, just transport it with Edge Snowball is the right way.

upvoted 1 times

Palee 1 month, 2 weeks ago

Selected Answer: B

D is close to correct but B makes more sense

upvoted 1 times

HeidiWan 1 month, 2 weeks ago

Selected Answer: B

The best option to save bandwidth

upvoted 1 times

melvis8 2 months, 3 weeks ago

Selected Answer: B

AWS snowball is ideal to transfer TB-PB of data from an on-prem connection to AWS cloud with little amount of internet connection required

upvoted 1 times

MGKYAING 5 months, 2 weeks ago

Selected Answer: B

AWS Snowball Edge is specifically designed for scenarios requiring the transfer of large datasets (terabytes or petabytes) with minimal network bandwidth usage.
It is ideal for one-time, large-scale migrations like the 70 TB of video files in this case.
For ongoing, smaller-scale data transfers, solutions like S3 File Gateway or AWS DataSync may be more appropriate.

upvoted 1 times

Sjb_009 5 months, 3 weeks ago

Selected Answer: B

The least possible network bandwidth says it all

upvoted 2 times

OmarRefaat 7 months, 3 weeks ago

B is Correct Answer

upvoted 1 times

Andreshere 8 months, 3 weeks ago

Selected Answer: B

The question states that the storage is no longer growing. This implies that we don't need to make any kind of data synchronization. Additionally, the total storage is 70 TB, which is a large amount of data. This implies high transfer costs. So, we can discard A, C and D options.
Correct option: A.
A Snowball device is a physical storage device which supports large data transfers. It is commonly used for transporting huge amounts of data from on-premises to AWS. Concretely, Snowball Edge is suitable for data transfers up to 80 TB. The transport times are between 1 and 2 weeks, so in case that we have hundreds of terabytes of data, we get them earlier than using Internet. In case that we need to transfer petabytes of data, it is recommended to use AWS Snowmobile, which is a physical track that transports data, up to 10 PB.

upvoted 2 times

Andreshere 1 year, 5 months ago

The correct answer is B not A, i misswrite that.

upvoted 2 times

Sandy4v 8 months, 3 weeks ago

Selected Answer: B

B is the correct answer

upvoted 1 times

KerasHanog 8 months, 4 weeks ago

I choose B
The key is:
- The total storage is 70 TB and is no longer growing.
- Using the least possible network bandwidth.

No longer Growing mean 1 time migrate, no need file gw
Least Possible Nw Bandwith --> Snowball Edge

upvoted 2 times

PaulGa 10 months, 1 week ago

PaulGa 10 months, 1 week ago

Ans B. Agree: Snowball Edge is designed for these types of operations; its more robust and secure because the operation is completed (relatively) quickly. Ans C doesn't really fly.

upvoted 1 times

Johnoppong101 10 months, 1 week ago

Select answer: B
Why? The essence of the S3 File Gateway is to provide a seamless interface for on-premises apps to store and retrieve data in Amazon S3 using standard protocols such as NFS and SMB. On the other hand,
As I write this, the first AWS Official use case for Snowball is to migrate data especially when network conditions are limited.

The question is a bit tricky. But applying simple logical linguistic analysis, "as soon as possible" coupled with "the least network bandwidth possible" means the question's focal point is network bandwidth. So, whatever the least network bandwidth is, it`s corresponding time to get the data into AWS S3 bucket is the value for "as soon as possible".

upvoted 3 times

Supriya_T 10 months, 2 weeks ago

Selected Answer: B

Snowball transfers data faster than the internet, and in this case, as the size of the data is large, so this would be the best option

upvoted 1 times

PR5577 10 months, 3 weeks ago

Selected Answer: B

Question specifically mentions minimal use of Network bandwidth. Storage gateway are mostly for using cloud storage on-premises. Usually data is copied one time using DataSync or other services like Snow devices.

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #7 *Topic 1*

A company has an application that ingests incoming messages. Dozens of other applications and microservices then quickly consume these messages. The number of messages varies drastically and sometimes increases suddenly to 100,000 each second. The company wants to decouple the solution and increase scalability.
Which solution meets these requirements?

A. Persist the messages to Amazon Kinesis Data Analytics. Configure the consumer applications to read and process the messages.

B. Deploy the ingestion application on Amazon EC2 instances in an Auto Scaling group to scale the number of EC2 instances based on CPU metrics.

C. Write the messages to Amazon Kinesis Data Streams with a single shard. Use an AWS Lambda function to preprocess messages and store them in Amazon DynamoDB. Configure the consumer applications to read from DynamoDB to process the messages.

D. Publish the messages to an Amazon Simple Notification Service (Amazon SNS) topic with multiple Amazon Simple Queue Service (Amazon SQS) subscriptions. Configure the consumer applications to process the messages from the queues. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (85%) Other (15%)

Comments

rein_chau Highly Voted 2 years, 8 months ago

Selected Answer: D

D makes more sense to me.

upvoted 56 times

daizy 2 years, 4 months ago

D. Publish the messages to an Amazon Simple Notification Service (Amazon SNS) topic with multiple Amazon Simple Queue Service (Amazon SQS) subscriptions. Configure the consumer applications to process the messages from the queues.

This solution uses Amazon SNS and SQS to publish and subscribe to messages respectively, which decouples the system and enables scalability by allowing multiple consumer applications to process the messages in parallel. Additionally, using Amazon

SQS with multiple subscriptions can provide increased resiliency by allowing multiple copies of the same message to be processed in parallel.

upvoted 18 times

SilentMilli 2 years, 5 months ago

By default, an SQS queue can handle a maximum of 3,000 messages per second. However, you can request higher throughput by contacting AWS Support. AWS can increase the message throughput for your queue beyond the default limits in increments of 300 messages per second, up to a maximum of 10,000 messages per second.

It's important to note that the maximum number of messages per second that a queue can handle is not the same as the maximum number of requests per second that the SQS API can handle. The SQS API is designed to handle a high volume of requests per second, so it can be used to send messages to your queue at a rate that exceeds the maximum message throughput of the queue.

upvoted 18 times

9014 2 years, 6 months ago

of course, the answer is D

upvoted 3 times

PhucVuu Highly Voted 8 months, 3 weeks ago

Selected Answer: D

Keywords:
- The number of messages varies drastically
- Sometimes increases suddenly to 100,000 each second

A: Incorrect - Don't confuse between Kinesis Data Analytics and Kinesis Data Stream =)) Kinesis Data Analytics will get the data from Kinesis Data Stream or Kinesis Data FireHose or MSK (Managed Stream for apache Kafka) for analytic purpose. It can not consume message and send to applications.
B: Incorrect - Base on the keywords -> Auto Scaling group not scale well because it need time to check the CPU metric and need time to start up the EC2 and the messages varies drastically. Example: we have to scale from 10 to 100 EC2. Our servers may be down a while when it was scaling.
C: Incorrect - Kinesis Data Streams can handle this case but we should increase the more shards but not single shard.
D: Correct: We can handle high workload well with fan-out pattern SNS + multiple SQS -> This is good for use case:
- The number of messages varies drastically
- Sometimes increases suddenly to 100,000 each second

upvoted 22 times

shinejh0528 2 years, 2 months ago

oh... I confused between Kinesis Data Analytics and Kinesis Data Stream as you mentioned... I solved several this type of questions, but SNS is always about 'notification', so i choose A. but i think Kinesis Data Analytics is just wrong, so D is most correct answer.

upvoted 2 times

clfigueiredo12 Most Recent 3 weeks, 2 days ago

Selected Answer: D

Letra D!

upvoted 1 times

vovanbi94 1 month, 1 week ago

Selected Answer: D

D the best choose

upvoted 1 times

Juju66 1 month, 3 weeks ago

Selected Answer: D

The other options are not correct.
Like Option A: Kinesis Data Analytics is designed for real-time data processing and analytics. Not for message queuing and decoupling.

upvoted 1 times

Aychi 2 months, 3 weeks ago

Selected Answer: D

D is correct answer

upvoted 1 times

melvis8 2 months, 3 weeks ago

Selected Answer: D

using an SQS queues to decouple this soluton will be the best approach since an SQS queue is very scalable and the number of messages to scale can be incremented by the user manually or by the use of security groups

upvoted 1 times

chente_diaz 3 months, 2 weeks ago

Selected Answer: D

I choose D because the question say "decouple/scalable" and the best services for that are SNS/SQS.

upvoted 1 times

Mrigraj12 4 months, 2 weeks ago

Selected Answer: D

D should be the answer as it is achieving a good decoupling:
1. Use SNS to create topics so messages for different applications will go on different topics
2. Create multiple SQS queues and subscribe to different SNS topics and let the different applications poll from their related SQS qeueus.

upvoted 1 times

MGKYAING 5 months, 2 weeks ago

Selected Answer: D

Amazon SNS + SQS provides a scalable and decoupled architecture:
SNS is a publish-subscribe messaging service that allows the ingestion application to broadcast messages to multiple subscribers.
SQS acts as the subscriber to the SNS topic, creating separate message queues for each consumer application. This ensures:
Scalability: Messages are distributed across queues, decoupling the producer and consumers.
Reliability: SQS queues store messages until consumer applications can process them, accommodating sudden spikes in message volume.

upvoted 1 times

hopefully2022 6 months, 2 weeks ago

Selected Answer: D

D is correct, and the easiest way to choose this answer choice is by the keyword "decouple." In order to decouple services, you can choose to use SQS/SNS (I believe there is a typo that says SOS)

upvoted 1 times

MGKYAING 7 months ago

The Correct Answer is D because it is required loosely couple and distributed queing system apporach.

upvoted 1 times

Buruguduystunstugudunstuy 8 months, 3 weeks ago

CORRECT ANSWER
The correct solution that meets these requirements is Option D: Publish the messages to an Amazon Simple Notification Service (Amazon SNS) topic with multiple Amazon Simple Queue Service (Amazon SQS) subscriptions. Configure the consumer applications to process the messages from the queues.

Option D involves using Amazon Simple Notification Service (SNS) and Amazon Simple Queue Service (SQS) to decouple the solution and increase scalability. SNS is a fully managed, publish-subscribe messaging service that allows you to send messages to multiple recipients simultaneously. SQS is a fully managed, distributed message queue service that enables you to store, process, and transmit messages between microservices, distributed systems, and serverless applications.

upvoted 4 times

Buruguduystunstugudunstuy 2 years, 5 months ago

To implement this solution, you would first publish the incoming messages to an SNS topic. You could then create multiple SQS subscriptions to the SNS topic, and configure the consumer applications to process the messages from the queues. This approach allows you to decouple the ingestion application from the consumer applications, and it allows you to scale the number of consumer applications independently of the ingestion application.

upvoted 4 times

Buruguduystunstugudunstuy 2 years, 5 months ago

WRONG AS EXPLAINED
Option A involves persisting the messages to Amazon Kinesis Data Analytics and configuring the consumer applications to read and process the messages. This option would not be the most efficient solution, as it would require the consumer applications to continuously poll Kinesis Data Analytics for new messages, which could impact their performance.

Option B involves deploying the ingestion application on Amazon EC2 instances in an Auto Scaling group to scale the number of EC2 instances based on CPU metrics. This option would not decouple the solution and increase scalability, as the consumer applications would still be directly connected to the ingestion application.

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option C involves writing the messages to Amazon Kinesis Data Streams with a single shard and using an AWS Lambda function to preprocess the messages and store them in DynamoDB. This option would not be the most efficient solution, as it would require the consumer applications to continuously poll DynamoDB for new messages, which could impact their performance.

Overall, Option D is the most effective solution for decoupling the solution and increasing scalability.

upvoted 5 times

karloscetina007 8 months, 3 weeks ago

Selected Answer: A

SNS and SQS still have an standard limit of tails under 3000 messages per sec, because SNS and SQS still have an standard limit of tails under 3000 messages per sec. It does not accomplishes with the requirement.

Perharps it's a possible solution: Amazon Kinesis Data Analytics. Configure the consumer applications to read and process the messages, but it requires that change the arch of this app, but this point is no matter on ther requirement.

upvoted 1 times

Fresbie99 1 year, 10 months ago

thats for FIFO queue.

upvoted 1 times

Fresbie99 1 year, 10 months ago

for standard queue there is nearly no limit to messages

upvoted 2 times

cookieMr 8 months, 3 weeks ago

Selected Answer: D

Option A: It is more suitable for real-time analytics and processing of streaming data rather than decoupling and scaling message ingestion and consumption.

Option B: It may help with scalability to some extent, but it doesn't provide decoupling.

Option C: It is a valid option, but it lacks the decoupling aspect. In this approach, the consumer applications would still need to read directly from DynamoDB, creating tight coupling between the ingestion and consumption processes.

Option D: It is the recommended solution for decoupling and scalability. The ingestion application can publish messages to an SNS topic, and multiple consumer apps can subscribe to the relevant SQS queues. SNS ensures that each message is delivered to all subscribed queues, allowing the consuming apps to independently process the messages at their own pace and scale horizontally as needed. This provides loose coupling, scalability, and fault tolerance, as the queues can handle message spikes and manage the consumption rate based on the consumer's processing capabilities.

upvoted 4 times

Guru4Cloud 8 months, 3 weeks ago

Selected Answer: D

The answer is D. Publish the messages to an Amazon Simple Notification Service (Amazon SNS) topic with multiple Amazon Simple Queue Service (Amazon SQS) subscriptions. Configure the consumer applications to process the messages from the queues.
This solution is the most scalable and decoupled solution for the given scenario. Amazon SNS is a pub/sub messaging service

that can be used to decouple applications. Amazon SQS is a fully managed message queuing service that can be used to store and process messages.
The solution would work as follows:
The ingestion application would publish the messages to an Amazon SNS topic.
The Amazon SNS topic would have multiple Amazon SQS subscriptions.
The consumer applications would subscribe to the Amazon SQS queues.
The consumer applications would process the messages from the Amazon SQS queues.

upvoted 2 times

Stevey 8 months, 3 weeks ago

D. is the answer.
The question states that there are dozens of other applications and microservices that consume these messages and that the volume of messages can vary drastically and increase suddenly. Therefore, you need a solution that can handle a high volume of messages, distribute them to multiple consumers, and scale quickly. SNS with SQS provides these capabilities.

Publishing messages to an SNS topic with multiple SQS subscriptions is a common AWS pattern for achieving both decoupling and scalability in message-driven systems. SNS allows messages to be fanned out to multiple subscribers, which in this case would be SQS queues. Each consumer application could then process messages from its SQS queue at its own pace, providing scalability and ensuring that all messages are processed by all consumer applications.

A. Amazon Kinesis Data Analytics is primarily used for real-time analysis of streaming data. It's not designed to distribute messages to multiple consumers.

upvoted 3 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #8 *Topic 1*

A company is migrating a distributed application to AWS. The application serves variable workloads. The legacy platform consists of a primary server that coordinates jobs across multiple compute nodes. The company wants to modernize the application with a solution that maximizes resiliency and scalability.
How should a solutions architect design the architecture to meet these requirements?

A. Configure an Amazon Simple Queue Service (Amazon SQS) queue as a destination for the jobs. Implement the compute nodes with Amazon EC2 instances that are managed in an Auto Scaling group. Configure EC2 Auto Scaling to use scheduled scaling.

B. Configure an Amazon Simple Queue Service (Amazon SQS) queue as a destination for the jobs. Implement the compute nodes with Amazon EC2 instances that are managed in an Auto Scaling group. Configure EC2 Auto Scaling based on the size of the queue. Most Voted

C. Implement the primary server and the compute nodes with Amazon EC2 instances that are managed in an Auto Scaling group. Configure AWS CloudTrail as a destination for the jobs. Configure EC2 Auto Scaling based on the load on the primary server.

D. Implement the primary server and the compute nodes with Amazon EC2 instances that are managed in an Auto Scaling group. Configure Amazon EventBridge (Amazon CloudWatch Events) as a destination for the jobs. Configure EC2 Auto Scaling based on the load on the compute nodes.

Correct Answer: *B*

Community vote distribution

B (92%) Other (8%)

Comments

rein_chau Highly Voted 2 years, 8 months ago

Selected Answer: B

A - incorrect: Schedule scaling policy doesn't make sense.
C, D - incorrect: Primary server should not be in same Auto Scaling group with compute nodes.
B is correct.

upvoted 88 times

Sinaneos Highly Voted 2 years, 8 months ago

Selected Answer: B

The answer seems to be B for me:
A: doesn't make sense to schedule auto-scaling
C: Not sure how CloudTrail would be helpful in this case, at all.
D: EventBridge is not really used for this purpose, wouldn't be very reliable

upvoted 24 times

K_SAA Most Recent 1 month ago

Selected Answer: B

keywords
resilient and scalability application
B is the correct answer since we need sqs designed as a destination to consume jobs from mutiple compute notes and use auto scaling group based on the size of the queue

A: is not considerd to be a correct answer since if we use scheduled scaling its not optimize in this use case if suddenly the workload unpredictable
C: cloudtrail is used for monitoring actions in aws not use for managing the workload
D:

upvoted 1 times

alfteezy91 2 months, 3 weeks ago

Selected Answer: B

Decouple with Amazon SQS and EC2 instances can be managed in an auto scaling group for scalability
https://docs.aws.amazon.com/prescriptive-guidance/latest/modernization-mainframe-decoupling-patterns/queue.html

upvoted 1 times

VitMoreira 3 months ago

Selected Answer: D

B doesn't make sense to me because you're scaling based on the queue size and not the requirements of the varying workloads.
Eventbridge can act as a trigger for workloads, so it might just cut, although it wouldn't be my first choice.

upvoted 1 times

MGKYAING 5 months, 2 weeks ago

Selected Answer: B

1.Amazon SQS for Decoupling:
Amazon SQS provides a fully managed message queue to decouple the primary server from the compute nodes. Jobs are sent to the queue, and compute nodes process them independently.
This architecture eliminates a single point of failure (the primary server) and increases resilience.

2.Auto Scaling Group for Compute Nodes:
EC2 instances managed in an Auto Scaling group process the jobs in the SQS queue. Auto Scaling dynamically adjusts the number of instances based on the queue size, ensuring scalability to handle variable workloads.

3.Scalability Based on Queue Size:
Scaling based on the size of the SQS queue ensures that the system adjusts to workload demands efficiently. When the queue grows, more instances are launched; when the queue shrinks, instances are terminated.

upvoted 2 times

PhucVuu 8 months, 3 weeks ago

Selected Answer: B

keywords:
- Legacy platform consists of a primary server that coordinates jobs across multiple compute nodes.
- Maximizes resiliency and scalability.

A: Incorrect - the question don't mention about schedule for high workload. So we don't use scheduled scaling for this case.
B: Correct - SQS can keep your message in the queue in case of high workload and if it too high we can increase the EC2 instance base on size of the queue.
C: Incorrect - AWS CloudTrail is API logs it is use for audit log of AWS user activity.
D: Incorrect - Event Bridge is use for filter event and trigger event.

upvoted 14 times

gx2222 8 months, 3 weeks ago

Selected Answer: B

-

B.
Explanation:
To maximize resiliency and scalability, the best solution is to use an Amazon SQS queue as a destination for the jobs. This decouples the primary server from the compute nodes, allowing them to scale independently. This also helps to prevent job loss in the event of a failure.

Using an Auto Scaling group of Amazon EC2 instances for the compute nodes allows for automatic scaling based on the workload. In this case, it's recommended to configure the Auto Scaling group based on the size of the Amazon SQS queue, which is a better indicator of the actual workload than the load on the primary server or compute nodes. This approach ensures that the application can handle variable workloads, while also minimizing costs by automatically scaling up or down the compute nodes as needed.

upvoted 5 times

Andreshere 8 months, 3 weeks ago

Selected Answer: B

SQS helps to process messages in case of variable workloads. The compute nodes must be implemented using EC2 instances (or alternatively, ECS tasks or managed Kubernetes nodes, but this option is not available). AutoScaling must be based on the workload, which is controlled by the queue. So, the correct option is B.
A is not correct because the instances should not scale based on a schedule which is not deterministic. On the contrary, scales based on the workload (queue size) is more effective.
AWS CloudTrail should not be used as a destination job and it is not related to the question. The same applies to EventBridge.

upvoted 3 times

WMF0187 8 months, 3 weeks ago

B:

Explanation:
Amazon SQS provides a reliable, highly scalable, and fully managed message queuing service that enables you to decouple and coordinate the components of a distributed application.
EC2 Auto Scaling allows you to automatically adjust the number of EC2 instances based on demand, ensuring that your application can handle variable workloads efficiently.
Auto Scaling based on the size of the queue ensures that your application scales out when there are many jobs to process and scales in when the job load decreases, providing cost efficiency and responsiveness to workload changes.

upvoted 2 times

abdulghaffar 11 months ago

According to them , Option C is correct how?

upvoted 1 times

PaulGa 8 months, 3 weeks ago

A lot of answer B's...
but I'm not convinced its Ans B which states: “Configure EC2 Auto Scaling based on the size of the queue” – because basing scaling on the size of the queue ignores the specific workload each job requires. The problem states “The application serves variable workloads” – you can’t determine the processing required for a variable workload based solely on queue size; this can only be done when you scope the size of the specific variable load – and that to my mind points to answer D: “Configure EC2 Auto Scaling based on the load on the compute nodes” – but then I run into the (potential) problem that Eventbridge may not be up to the task…

upvoted 2 times

Zwein 9 months ago

why are almost all of the "correct answers" I see on this site all wrong? how the fuck is this an educational resource? good thing the community voting system exists or else this side would be pure unadulterated putrid shit.

upvoted 2 times

LeonDong 9 months, 2 weeks ago

Selected Answer: B

B for sure

upvoted 1 times

ChymKuBoy 1 year ago

Selected Answer: B

B for sure

upvoted 1 times

OBIOHAnze 1 year ago

Selected Answer: D

option D leverages serverless services (EventBridge) and Auto Scaling for a modern, scalable, and resilient architecture suitable for the distributed application with varying workloads.

upvoted 2 times

lixep 1 year ago

Selected Answer: C

In option C, Ignore the line that's talking about cloud trail and then the answer would make much more sense.

upvoted 4 times

lofzee 1 year, 1 month ago

Selected Answer: B

B for sure

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #9 *Topic 1*

A company is running an SMB file server in its data center. The file server stores large files that are accessed frequently for the first few days after the files are created. After 7 days the files are rarely accessed.
The total data size is increasing and is close to the company's total storage capacity. A solutions architect must increase the company's available storage space without losing low-latency access to the most recently accessed files. The solutions architect must also provide file lifecycle management to avoid future storage issues.
Which solution will meet these requirements?

A. Use AWS DataSync to copy data that is older than 7 days from the SMB file server to AWS.

B. Create an Amazon S3 File Gateway to extend the company's storage space. Create an S3 Lifecycle policy to transition the data to S3 Glacier Deep Archive after 7 days. **Most Voted**

C. Create an Amazon FSx for Windows File Server file system to extend the company's storage space.

D. Install a utility on each user's computer to access Amazon S3. Create an S3 Lifecycle policy to transition the data to S3 Glacier Flexible Retrieval after 7 days.

Correct Answer: *B*

Community vote distribution

B (85%) Other (15%)

Comments

Sinaneos Highly Voted 8 months, 3 weeks ago

Answer directly points towards file gateway with lifecycles,

D is wrong because utility function is vague and there is no need for flexible storage.

upvoted 54 times

gianola 10 months, 2 weeks ago

I think B is wrong because they are not asking you to add more room, they are asking you to have more room available.

upvoted 1 times

Udoyen 2 years, 6 months ago

Yes it might be vague but how do we keep the low-latency access that only flexible can offer?

upvoted 5 times

SuperDuperPooperScooper 1 year, 10 months ago

Low-latency access is only required for the first 7 days, B maintains that fast access for 7 days and only then are the files sent to Glacier Archive

upvoted 3 times

Nava702 1 year, 9 months ago

It says low-latency is required for the most recently accessed files, not new ones. So if an older file is retrieved from deep archive, it should then readily be accessible, according to the question, which points toward Flexible retrieval. However the utility portion in the answer D is vague.

upvoted 4 times

francisizme 3 months, 1 week ago

D will require users to change how they access files, while B only needs the SMB server to mount the gateway, and then the server can read and write files to S3 just like a normal network drive.

upvoted 1 times

javitech83 Highly Voted 8 months, 3 weeks ago

Selected Answer: B

B answwer is correct. low latency is only needed for newer files. Additionally, File GW provides low latency access by caching frequently accessed files locally so answer is B

upvoted 36 times

39sachin Most Recent 1 month ago

Selected Answer: B

Configured S3 buckets are accessible using the NFS and SMB protocol
Most recently used data is cached in the file gateway

upvoted 1 times

ak240519 4 months ago

Selected Answer: B

File Gateway helps with lower latency and the files after 7 days should be moved to Glacier as they are no longer being accessed.

upvoted 1 times

adamatic 4 months, 1 week ago

Selected Answer: B

B extends the storage with s3 gateway and meets lifecycle policy requirements

upvoted 1 times

Sandy4v 8 months, 2 weeks ago

B is correct

upvoted 1 times

HayLLlHuK 8 months, 3 weeks ago

The same question and answer explanation exists in a Udemy course.
Correct answer is B.
Amazon S3 File Gateway provides a seamless way to connect to the cloud to store application data files and backup images as durable objects in Amazon S3 cloud storage. Amazon S3 File Gateway offers SMB or NFS-based access to data in Amazon S3 with local caching.
It can be used for on-premises data-intensive Amazon EC2-based applications that need file protocol access to S3 object storage.
Lifecycle policies can then transition the data to S3 Glacier Deep Archive after 7 days.

D is wrong because is involves too much extra configuration which is unnecessary.

upvoted 5 times

Buruguduystunstugudunstuy 8 months, 3 weeks ago

Selected Answer: B

Option B, creating an Amazon S3 File Gateway and an S3 Lifecycle policy to transition data to S3 Glacier Deep Archive, would meet the requirements specified in the prompt.

The S3 File Gateway allows you to store and retrieve objects in Amazon S3 using standard file system protocols, such as SMB and NFS. This would provide additional storage space for the company's data and allow for low-latency access to the most recently accessed files, as the data would still be stored on the SMB file server.

upvoted 6 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option A, using AWS DataSync to copy data that is older than 7 days from the SMB file server to AWS, would not provide additional storage space for the company's data and would not allow for low-latency access to the most recently accessed files.

Option C, creating an FSx for Windows File Server file system, would provide additional storage space but would not include file lifecycle management.

Option D, installing a utility on each user's computer to access Amazon S3 and creating an S3 Lifecycle policy, would not provide additional storage space on the company's file server.

upvoted 5 times

gx2222 8 months, 3 weeks ago

Selected Answer: B

Explanation:

Since the company needs to increase available storage space while maintaining low-latency access to recently accessed files and implement file lifecycle management to avoid future storage issues, the best solution is to use Amazon S3 with a File Gateway.

Using an Amazon S3 File Gateway, the company can access its SMB file server through an S3 bucket. This provides low-latency access to recently accessed files by caching them on the gateway appliance. The solution also supports file lifecycle management by using S3 Lifecycle policies to transition files to lower cost storage classes after they haven't been accessed for a certain period of time.

In this case, the company can create an S3 Lifecycle policy to transition files to S3 Glacier Deep Archive after 7 days of not being accessed. This would allow the company to store large amounts of data at a lower cost, while still having easy access to recently accessed files.

upvoted 3 times

PhucVuu 8 months, 3 weeks ago

Selected Answer: B

Keywords:
- After 7 days the files are rarely accessed.
-The total data size is increasing and is close to the company's total storage capacity.
- Increase the company's available storage space without losing low-latency access to the most recently accessed files. -> (for rarely accessed files we can access it with high-latency)
- Must also provide file lifecycle management to avoid future storage issues.

A: Incorrect - Don't mention how to increase company's available storage space.
B: Correct - extend storage space and fast access with S3 File Gateway (cache recent access file), reduce cost and storage by move to S3 Glacier Deep Archive after 7 days.
C: Incorrect - Didn't handle file lifecycle management.
D: Incorrect - Don't mention about increase the company's available storage space.

upvoted 12 times

McLobster 8 months, 3 weeks ago

Selected Answer: A

according to documentation the minimum storage timeframe for an object inside S3 before being able to transition using lifecycle policy is 30 days , so those 7 days policies kinda seem wrong to me

Transition actions – These actions define when objects transition to another storage class. For example, you might choose to transition objects to the S3 Standard-IA storage class 30 days after creating them, or archive objects to the S3 Glacier Flexible Retrieval storage class one year after creating them. For more information, see Using Amazon S3 storage classes.

I was thinking of option A using DataSync as a scheduled task? am i wrong here?

upvoted 2 times

Andreshere 8 months, 3 weeks ago

Andreshere 8 months, 3 weeks ago

A. DataSync is focused on transferring data when we need synchronization (for example, from an on-premises DB that updates daily to a DB in AWS). In this case, we don't need to transfer data or synchronize data, we only need to increase the storage. So, this option is not correct.
B. S3 File Gateway allows communication between a File System/Server and S3, and it supports SMB protocol. We can use S3 FGW to move files from the FS to the S3 bucket. Then, the question says that files older than seven days are rarely accessed, so we can transition to S3 Glacier for those files to archive, in a costly-efficient way. So, this option is correct.
C. You have two different storages for saving the files, making the interoperability unnecessarily more complex (and you need constant data synchronization, regarding to option A).
D. This is a mess. The solution is not scalable and It depends on the number of users, which is not a static number. Additionally, Flexible Retrieval is unnecessary.

upvoted 5 times

akshay243007 8 months, 3 weeks ago

Selected Answer: B

answwer is correct. low latency is only needed for newer files. Additionally, File GW provides low latency access by caching frequently accessed files locally so answer is B

upvoted 1 times

CHM3 9 months, 3 weeks ago

Selected Answer: B

B is correct

upvoted 1 times

PaulGa 10 months, 1 week ago

Ans B: Low latency is only required for recent files in last 7 days; the rest can effectively be archived.

upvoted 1 times

Sebbster 10 months, 1 week ago

I'm new to this site, B is obviously correct, so how can D be shown as the "Correct Answer"? Makes no sense..

upvoted 2 times

WMF0187 11 months ago

B:

Explanation:
Amazon S3 File Gateway: This service provides a seamless and secure integration between on-premises environments and the Amazon S3 cloud storage, allowing users to store and retrieve objects in S3 using the standard file protocols.
S3 Lifecycle policy: By creating a lifecycle policy, you can automatically transition data that is older than 7 days to a more cost-effective storage class like S3 Glacier Deep Archive, which is suitable for infrequently accessed data.

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #10 *Topic 1*

A company is building an ecommerce web application on AWS. The application sends information about new orders to an Amazon API Gateway REST API to process. The company wants to ensure that orders are processed in the order that they are received.
Which solution will meet these requirements?

A. Use an API Gateway integration to publish a message to an Amazon Simple Notification Service (Amazon SNS) topic when the application receives an order. Subscribe an AWS Lambda function to the topic to perform processing.

B. Use an API Gateway integration to send a message to an Amazon Simple Queue Service (Amazon SQS) FIFO queue when the application receives an order. Configure the SQS FIFO queue to invoke an AWS Lambda function for processing.
Most Voted

C. Use an API Gateway authorizer to block any requests while the application processes an order.

D. Use an API Gateway integration to send a message to an Amazon Simple Queue Service (Amazon SQS) standard queue when the application receives an order. Configure the SQS standard queue to invoke an AWS Lambda function for processing.

Correct Answer: *B*

Community vote distribution

B (98%) A

Comments

Sinaneos Highly Voted 8 months, 3 weeks ago
Selected Answer: B

B because FIFO is made for that specific purpose
upvoted 68 times

rein_chau Highly Voted 2 years, 8 months ago
Selected Answer: B

Should be B because SQS FIFO queue guarantees message order.
upvoted 28 times

K_SAA Most Recent 1 month ago
Selected Answer: B

keywords: orders should be process in the order when it receives
B is correct answer, use API gateway intergation with FIFO SQS invoke lambda function for processing order
A; SNS + lambda: SNS does not grant order messages to subscribers
C: is not the correct one since we wanted to process the message in order so we do not have to block
D: using standard queue does not grant the order of messages being sent
upvoted 1 times

sdjfidhbgidfbivgv 2 months ago

Selected Answer: B

B SQS FIFO is made for that
upvoted 1 times

offek 3 months, 2 weeks ago

Selected Answer: B

FIFO make sure to send the data exactly in the same order
upvoted 1 times

ak240519 4 months ago

Selected Answer: B

B is the only solution that guarantees message order.
upvoted 1 times

adamatic 4 months, 1 week ago

Selected Answer: B

Has to be FIFO queue for this requirement
upvoted 1 times

Mrigraj12 4 months, 2 weeks ago

Selected Answer: B

SQS FIFO as if want to maintain order of messages recieved to order of messages sent. It is like QUEUE data structutre in DSA
upvoted 1 times

surez 4 months, 4 weeks ago

Selected Answer: B

SQS FIFO gurantees message order.
upvoted 1 times

MGKYAING 5 months, 2 weeks ago

Selected Answer: B

Explanation:
FIFO Queues: Amazon SQS FIFO queues are specifically designed to maintain the order of messages. This ensures that messages are processed in the exact sequence they were sent, which is crucial for order processing in an e-commerce application.

Why other options are less suitable:
A. Amazon SNS: SNS is a publish-subscribe service. It doesn't guarantee message order.
C. API Gateway Authorizer: Authorizers are used for authentication and authorization, not for managing message order.
D. Amazon SQS Standard Queue: Standard SQS queues do not guarantee message order.
upvoted 3 times

ondan 8 months, 1 week ago

Standard queues ensure at-least-once message delivery, but due to the highly distributed architecture, more than one copy of a message might be delivered, and messages may occasionally arrive out of order.
upvoted 3 times

Abishek016 8 months, 2 weeks ago

AnswerB - SQS FIFO queues ensure that messages are processed in the order they are received, which perfectly matches the requirement of maintaining order.
upvoted 2 times

paobalinas 8 months, 3 weeks ago

A lot of answers seem to not match the most voted. i'm confused which to follow.

upvoted 2 times

gx2222 8 months, 3 weeks ago

Selected Answer: B

Explanation:

To ensure that orders are processed in the order that they are received, the best solution is to use an Amazon SQS FIFO (First-In-First-Out) queue. This type of queue maintains the exact order in which messages are sent and received.

In this case, the application can send information about new orders to an Amazon API Gateway REST API, which can then use an API Gateway integration to send a message to an Amazon SQS FIFO queue for processing. The queue can then be configured to invoke an AWS Lambda function to perform the necessary processing on each order. This ensures that orders are processed in the exact order in which they are received.

upvoted 5 times

PhucVuu 8 months, 3 weeks ago

Selected Answer: B

Keywords:
- Orders are processed in the order that they are received.

A: Incorrect - SNS just for notification like send email, SMS. It don't retain the data in the queue and it's used pub-sub pattern.
B: Correct - SQS FIFO will help message process in order. FIFO -> first in first out.
C: Incorrect - with this solution we will create blocker app not good app =))
D: Incorrect - SQS standard don't guarantee the order.

upvoted 12 times

Guru4Cloud 8 months, 3 weeks ago

Selected Answer: B

Explanation:
- Amazon API Gateway will be used to receive the orders from the web application.
- Instead of directly processing the orders, the API Gateway will integrate with an Amazon SQS FIFO queue.
- FIFO (First-In-First-Out) queues in Amazon SQS ensure that messages are processed in the order they are received.
- By using a FIFO queue, the order processing is guaranteed to be sequential, ensuring that the first order received is processed before the next one.
- An AWS Lambda function can be configured to be triggered by the SQS FIFO queue, processing the orders as they arrive

upvoted 6 times

Andreshere 8 months, 3 weeks ago

Selected Answer: B

A. This option could be correct if we use SNS FIFO option, but this is not the case stated in the question. Additionally, this task is more efficient done by a queue than a subscription service. So, this option is not correct.
B. We use a queue, which is efficient processing messages. Additionally, it preserves the order since it is a FIFO queue. Meanwhile we don’t have any kind of messages throughput limitation, this option is correct.
C. This option discards any messages that are still processing, which is not a good solution.
D. Same option as B but using a normal queue, which does not preserve the order. Incorrect.

upvoted 6 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #11 *Topic 1*

A company has an application that runs on Amazon EC2 instances and uses an Amazon Aurora database. The EC2 instances connect to the database by using user names and passwords that are stored locally in a file. The company wants to minimize the operational overhead of credential management.
What should a solutions architect do to accomplish this goal?

A. Use AWS Secrets Manager. Turn on automatic rotation. **Most Voted**

B. Use AWS Systems Manager Parameter Store. Turn on automatic rotation.

C. Create an Amazon S3 bucket to store objects that are encrypted with an AWS Key Management Service (AWS KMS) encryption key. Migrate the credential file to the S3 bucket. Point the application to the S3 bucket.

D. Create an encrypted Amazon Elastic Block Store (Amazon EBS) volume for each EC2 instance. Attach the new EBS volume to each EC2 instance. Migrate the credential file to the new EBS volume. Point the application to the new EBS volume.

Correct Answer: *A*

Community vote distribution

A (96%) B

Comments

Sinaneos Highly Voted 2 years, 8 months ago

Selected Answer: A

B is wrong because parameter store does not support auto rotation, unless the customer writes it themselves, A is the answer.
upvoted 102 times

17Master 2 years, 7 months ago

READ!!! AWS Secrets Manager is a secrets management service that helps you protect access to your applications, services, and IT resources. This service enables you to rotate, manage, and retrieve database credentials, API keys, and other secrets throughout their lifecycle.
upvoted 30 times

hro 1 year, 2 months ago

A - additionally, Aurora manages the settings for the secret and rotates the secret every seven days by default.
upvoted 3 times

iCcma 2 years, 7 months ago

ty bro, I was confused about that and you just mentioned the "key" phrase, B doesn't support autorotation

upvoted 3 times

cookieMr Highly Voted 1 year, 12 months ago

Selected Answer: A

Option A: Using AWS Secrets Manager and enabling automatic rotation is the recommended solution for minimizing the operational overhead of credential management. AWS Secrets Manager provides a secure and centralized service for storing and managing secrets, such as database credentials. By leveraging Secrets Manager, the application can retrieve the database credentials programmatically at runtime, eliminating the need to store them locally in a file. Enabling automatic rotation ensures that the database credentials are regularly rotated without manual intervention, enhancing security and compliance.

upvoted 8 times

Kamatchi Most Recent 2 months, 2 weeks ago

Selected Answer: A

This is asked in the exam I have taken today

upvoted 2 times

MundiChor 3 months, 2 weeks ago

Selected Answer: A

Order of elimination:
B. Parameter Store can store secrets. You can enforce TTL to expire credentials but you cannot auto rotate
C and D are plan overhead and not secure enough.

upvoted 1 times

ak240519 4 months ago

Selected Answer: A

A is the correct answer. as Instance store doesn't support auto rotation

upvoted 1 times

adamatic 4 months, 1 week ago

Selected Answer: A

AWS Secrets Manager is a service that helps users manage, rotate, and retrieve secrets for applications, services, and IT resources. It can be used to secure secrets for AWS Cloud, third-party services, and on-premises
-Automatic rotation: Secrets can be automatically rotated

upvoted 1 times

MGKYAING 5 months, 2 weeks ago

Selected Answer: A

AWS Secrets Manager:
Is a managed service specifically designed to securely store and retrieve secrets, such as database credentials, API keys, and SSH keys.
Provides features like automatic rotation, which helps to reduce the risk of compromised credentials.
Integrates seamlessly with many AWS services, including Amazon RDS (which Aurora is a part of).

upvoted 2 times

Gizmo2022 7 months ago

Answer is A
https://aws.amazon.com/cn/blogs/security/how-to-connect-to-aws-secrets-manager-service-within-a-virtual-private-cloud/
https://aws.amazon.com/blogs/security/rotate-amazon-rds-database-credentials-automatically-with-aws-secrets-manager/

upvoted 1 times

Abishek016 8 months, 2 weeks ago

Selected Answer: A

This is an ideal solution. Secrets Manager can rotate credentials automatically and ensures that the EC2 instances retrieve the most recent credentials securely.

upvoted 3 times

Buruguduystunstugudunstuy 8 months, 3 weeks ago

Buruguduystunstugudunstuy 3 months, 3 weeks ago

Selected Answer: A

Option A, using AWS Secrets Manager and turning on automatic rotation, would be the best solution to minimize the operational overhead of credential management.

AWS Secrets Manager is a service that makes it easier to manage secrets, such as database credentials, by storing and rotating them automatically. By turning on automatic rotation, you can ensure that the secrets are regularly rotated, reducing the risk of unauthorized access to the database. This would minimize the operational overhead of credential management, as you would not have to manually rotate the secrets or update the EC2 instances with the new credentials.

upvoted 3 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option B, using AWS Systems Manager Parameter Store and turning on automatic rotation, would not be suitable for storing secrets, such as database credentials, as it is intended for storing system parameters.

Option C, creating an S3 bucket to store objects that are encrypted with an AWS KMS encryption key and migrating the credential file to the S3 bucket, would not provide automatic rotation of the secrets.

Option D, creating an encrypted EBS volume and migrating the credential file to the new EBS volume, would not provide automatic rotation of the secrets.

upvoted 1 times

Ruffyit 8 months, 3 weeks ago

A: READ!!! AWS Secrets Manager is a secrets management service that helps you protect access to your applications, services, and IT resources. This service enables you to rotate, manage, and retrieve database credentials, API keys, and other secrets throughout their lifecycle.

It says SSM Parameter store cant rotate automatically.

upvoted 2 times

jallaix 8 months, 3 weeks ago

Everybody here voting A, but only the master user's password of the Aurora database can be automatically stored and rotated. Who uses the master user's credentials in their application ? It looks to me like a serious security issue...

Moreover answer A is not complete, missing steps are:
- create IAM role to get secret
- assign IAM role to EC2 instance
- adapt the application to retrieve the secret from Secrets Manager instead of erading the file
- make sure retrieval occurs every week
I dont' call that minimizing operational overhead... Answer D is a lot more simple.

In a real situation, none of these answers are relevant.

upvoted 3 times

iyiola_daniel 9 months, 3 weeks ago

Same thing I thought. Answer D seems simpler, but option A is the best approach.

upvoted 1 times

griggrig 9 months, 2 weeks ago

Selected Answer: A

Option A , because of leas overhead.

upvoted 1 times

parth_g_mehta 11 months ago

Selected Answer: A

Parameter Store: Storing and managing a database connection string or API endpoint URL that doesn’t require frequent rotation.

Secrets Manager: Storing and managing database credentials that need to be rotated regularly for security compliance.

upvoted 1 times

JalimRabeiBR 1 year ago

Answer A is correct
upvoted 1 times

OctavioBatera 1 year, 3 months ago

Selected Answer: A

Secrets Manager, as The Mandalorian would say "this is the way!"
upvoted 1 times

TilTil 1 year, 3 months ago

Selected Answer: A

SSM has no automatic rotation.
upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #12 *Topic 1*

A global company hosts its web application on Amazon EC2 instances behind an Application Load Balancer (ALB). The web application has static data and dynamic data. The company stores its static data in an Amazon S3 bucket. The company wants to improve performance and reduce latency for the static data and dynamic data. The company is using its own domain name registered with Amazon Route 53.
What should a solutions architect do to meet these requirements?

A. Create an Amazon CloudFront distribution that has the S3 bucket and the ALB as origins. Configure Route 53 to route traffic to the CloudFront distribution. **Most Voted**

B. Create an Amazon CloudFront distribution that has the ALB as an origin. Create an AWS Global Accelerator standard accelerator that has the S3 bucket as an endpoint Configure Route 53 to route traffic to the CloudFront distribution.

C. Create an Amazon CloudFront distribution that has the S3 bucket as an origin. Create an AWS Global Accelerator standard accelerator that has the ALB and the CloudFront distribution as endpoints. Create a custom domain name that points to the accelerator DNS name. Use the custom domain name as an endpoint for the web application.

D. Create an Amazon CloudFront distribution that has the ALB as an origin. Create an AWS Global Accelerator standard accelerator that has the S3 bucket as an endpoint. Create two domain names. Point one domain name to the CloudFront DNS name for dynamic content. Point the other domain name to the accelerator DNS name for static content. Use the domain names as endpoints for the web application.

Correct Answer: *A*

Community vote distribution

A (76%) C (24%)

Comments

Kartikey140 Highly Voted 8 months, 3 weeks ago

Answer is A
Explanation - AWS Global Accelerator vs CloudFront
• They both use the AWS global network and its edge locations around the world
• Both services integrate with AWS Shield for DDoS protection.
• CloudFront
• Improves performance for both cacheable content (such as images and videos)
• Dynamic content (such as API acceleration and dynamic site delivery)
• Content is served at the edge
• Global Accelerator

• Improves performance for a wide range of applications over TCP or UDP
• Proxying packets at the edge to applications running in one or more AWS Regions.
• Good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP
• Good for HTTP use cases that require static IP addresses
• Good for HTTP use cases that required deterministic, fast regional failover

upvoted 125 times

Mihailo34 12 months ago

A, adding to the excellent explanation by Kartikey140, the solution under C uses a custom DNS name, the question specifies: "The company is using its own domain name registered with Amazon Route 53"

upvoted 7 times

daizy 2 years, 4 months ago

By creating a CloudFront distribution that has both the S3 bucket and the ALB as origins, the company can reduce latency for both the static and dynamic data. The CloudFront distribution acts as a content delivery network (CDN), caching the data closer to the users and reducing the latency. The company can then configure Route 53 to route traffic to the CloudFront distribution, providing improved performance for the web application.

upvoted 19 times

kanweng Highly Voted 8 months, 3 weeks ago

Selected Answer: A

Q: How is AWS Global Accelerator different from Amazon CloudFront?

A: AWS Global Accelerator and Amazon CloudFront are separate services that use the AWS global network and its edge locations around the world. CloudFront improves performance for both cacheable content (such as images and videos) and dynamic content (such as API acceleration and dynamic site delivery). Global Accelerator improves performance for a wide range of applications over TCP or UDP by proxying packets at the edge to applications running in one or more AWS Regions. Global Accelerator is a good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP, as well as for HTTP use cases that specifically require static IP addresses or deterministic, fast regional failover. Both services integrate with AWS Shield for DDoS protection.

upvoted 42 times

Juju66 Most Recent 1 month, 3 weeks ago

Selected Answer: A

Option C is wrong , because:
This option introduces additional complexity and cost by using AWS Global Accelerator. CloudFront alone can handle the distribution of both static and dynamic content effectively.

upvoted 1 times

SamKuo 5 months, 3 weeks ago

Selected Answer: C

AWS Global Accelerator:Dynamic content
CloudFront:static content

upvoted 1 times

PhucVuu 8 months, 3 weeks ago

Keywords:
- The web application has static data and dynamic data. Static data in an Amazon S3 bucket.
- Improve performance and reduce latency for the static data and dynamic data.
- The company is using its own domain name registered with Amazon Route 53.
A: Correct - CloudFront has the Edge location and the cache for dynamic and static
B: Incorrect - AWS Global Accelerator don't have cache function, so static file need to be load directly from S3 every time.
- Beside that we configure CloudFront -> ALB, Accelerator -> S3, Route 53 -> CloudFront. It means that all the traffic go to CloudFront only, Acclerator don't have any traffic.
C: Incorrect - Global Accelerator can configure CloudFront as the endpoint.
D: Incorrect - We already have domain name. Why will we use new domain name? Will we change to new domain name? How everyone know you new domain name?

upvoted 15 times

zoe9z 6 months, 1 week ago

For C, it seems like Global Accelerator does not support CloudFront as an endpoint

upvoted 2 times

M0SHE 8 months, 3 weeks ago

Selected Answer: A

A. Create an Amazon CloudFront distribution that has the S3 bucket and the ALB as origins. Configure Route 53 to route traffic to the CloudFront distribution.

Here's the reasoning:

CloudFront with Multiple Origins: CloudFront allows you to set up multiple origins for your distribution, so you can use both the ALB (for dynamic content) and the S3 bucket (for static content) as origins. This means that both your dynamic and static content can be served through CloudFront, which will cache content at edge locations to reduce latency.
Route 53 Integration with CloudFront: Amazon Route 53 can be easily configured to route traffic for your domain to a CloudFront distribution. Users will access your domain, and Route 53 will direct them to the nearest CloudFront edge location.

upvoted 3 times

aropl 8 months, 3 weeks ago

A is correct, other answers have wrong origin or endpoint types.
Cloudfront supports multiple origins on the same distribution (ALB and S3) in our case.
B incorrect - Global Accelerator Standard accelerator doesn;t support s3 endpoints
c incorrect - Global Accelerator Standard accelerator doesn't support CloudFront distribution as endpoint
D incorrect - Global Accelerator Standard accelerator doesn't support s3 endpoints

upvoted 10 times

rainiverse 8 months, 3 weeks ago

Selected Answer: A

I'm wavering between A and C.
With dynamic content, CloudFront is cacheable and that's not good.
But with answer C, AWS Global doesn't support Cloudfront endpoint
"Endpoints for standard accelerators in AWS Global Accelerator can be Network Load Balancers, Application Load Balancers, Amazon EC2 instances, or Elastic IP addresses. "
So I choose A

upvoted 2 times

PaulGa 10 months ago

Selected Answer: C

Ans C - for the good reasons given by Diddy99. Altho' Ans A could do it, it is not the best optimised answer; Ans C is, but at cost of a custom domain name (which I don't like)

upvoted 1 times

KTEgghead 10 months, 3 weeks ago

Selected Answer: A

Cloudfront caches content at edge locations, reduces latency, and can serve static content from S3 buckets. It can also accelerate dynamic content from EC2. CloudFront maintains a persistent pool of connections to the origin, which minimises the overhead of establishing new connections. Any of these questions with "latency" and "improve performance" smell like CloudFront.

upvoted 1 times

creamymangosauce 11 months ago

Selected Answer: A

A - CloudFront for caching static content. No need for Global Accelerator since no static IP is required

upvoted 2 times

bishtr3 11 months ago

A

upvoted 1 times

Mihailo34 12 months ago

A, adding to the excellent explanation by Kartikey140, the solution under C uses a custom DNS name, the question specifies: "The company is using its own domain name registered with Amazon Route 53"

upvoted 1 times

diddy99 12 months ago

Selected Answer: C

Answer is C
Explanation:
A: Using Cloudfront to cache static content is perfect for low latency and performance. However, caching dynamic content from ALB through cloudfront might not be efficient as dynamic contents is often personalized and are not good for caching.
B: Using cloudfront to cache dynamic contents from ALB is not the most efficient approach
C: Using amazon cloudfront to cache the static data from S3 ensures efficient distribution of static contents globally. AWS Global accelerator routes traffic to the nearest AWS EDGE location. Hence, routing is optimized to both the ALB (Dynamic contents) and Cloud front distribution.

upvoted 6 times

ChymKuBoy 1 year ago

Selected Answer: A

A for sure

upvoted 1 times

OBIOHAnze 1 year ago

Selected Answer: A

By using CloudFront with separate origins for static and dynamic content, the company can achieve improved performance and reduced latency for both types of data. Route 53 then intelligently routes traffic based on the requested object, ensuring a smooth user experience.

upvoted 1 times

ManikRoy 1 year, 1 month ago

Selected Answer: A

It would have made sense to use S3 bucket as the origin for cloud front and ALB as the end point for global accelerator. However the option C messes it up when it mentions also the cloud front distribution as the end point for global accelerator standard (which is not supported). As this is not possible the only option left is A to use Cloud front for both S3 & ALB.

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #13 *Topic 1*

A company performs monthly maintenance on its AWS infrastructure. During these maintenance activities, the company needs to rotate the credentials for its Amazon RDS for MySQL databases across multiple AWS Regions.
Which solution will meet these requirements with the LEAST operational overhead?

A. Store the credentials as secrets in AWS Secrets Manager. Use multi-Region secret replication for the required Regions. Configure Secrets Manager to rotate the secrets on a schedule. **Most Voted**

B. Store the credentials as secrets in AWS Systems Manager by creating a secure string parameter. Use multi-Region secret replication for the required Regions. Configure Systems Manager to rotate the secrets on a schedule.

C. Store the credentials in an Amazon S3 bucket that has server-side encryption (SSE) enabled. Use Amazon EventBridge (Amazon CloudWatch Events) to invoke an AWS Lambda function to rotate the credentials.

D. Encrypt the credentials as secrets by using AWS Key Management Service (AWS KMS) multi-Region customer managed keys. Store the secrets in an Amazon DynamoDB global table. Use an AWS Lambda function to retrieve the secrets from DynamoDB. Use the RDS API to rotate the secrets.

Correct Answer: *A*

Community vote distribution

A (100%)

Comments

rein_chau Highly Voted 2 years, 8 months ago

Selected Answer: A

A is correct.

upvoted 24 times

PhucVuu Highly Voted 8 months, 3 weeks ago

Selected Answer: A

Keywords:
- rotate the credentials for its Amazon RDS for MySQL databases across multiple AWS Regions
- LEAST operational overhead

A: Correct - AWS Secrets Manager supports
- Encrypt credential for RDS, DocumentDb, Redshift, other DBs and key/value secret.
- multi-region replication

multi-region replication.
- Remote base on schedule
B: Incorrect - Secure string parameter only apply for Parameter Store. All the data in AWS Secrets Manager is encrypted
C: Incorrect - don't mention about replicate S3 across region.
D: Incorrect - So many steps compare to answer A =))

upvoted 14 times

MundiChor Most Recent 3 months, 2 weeks ago

Selected Answer: A

Capability to rotate secrets and tight coupling with RDS

upvoted 1 times

Vandaman 3 months, 3 weeks ago

Selected Answer: A

No other explanation necessary

upvoted 1 times

Mrigraj12 4 months, 2 weeks ago

Selected Answer: A

A is the right answer as it takes minutes to setup. so least operational head

upvoted 1 times

trinh_le 7 months, 1 week ago

Selected Answer: A

A: correct - the secret manager supports rotating credentials
B: incorrect - Parameter Store does not perform any cryptographic operations. Instead, it relies on AWS KMS to encrypt and decrypt secure string parameter values
C and D: incorrect - handles through Lambda, require more operational overhead

upvoted 1 times

SilentMilli 8 months, 3 weeks ago

Selected Answer: A

AWS Secrets Manager is a secrets management service that enables you to store, manage, and rotate secrets such as database credentials, API keys, and SSH keys. Secrets Manager can help you minimize the operational overhead of rotating credentials for your Amazon RDS for MySQL databases across multiple Regions. With Secrets Manager, you can store the credentials as secrets and use multi-Region secret replication to replicate the secrets to the required Regions. You can then configure Secrets Manager to rotate the secrets on a schedule so that the credentials are rotated automatically without the need for manual intervention. This can help reduce the risk of secrets being compromised and minimize the operational overhead of credential management.

upvoted 5 times

Buruguduystunstugudunstuy 8 months, 3 weeks ago

Selected Answer: A

Option A, storing the credentials as secrets in AWS Secrets Manager and using multi-Region secret replication for the required Regions, and configuring Secrets Manager to rotate the secrets on a schedule, would meet the requirements with the least operational overhead.

AWS Secrets Manager allows you to store, manage, and rotate secrets, such as database credentials, across multiple AWS Regions. By enabling multi-Region secret replication, you can replicate the secrets across the required Regions to allow for seamless rotation of the credentials during maintenance activities. Additionally, Secrets Manager provides automatic rotation of secrets on a schedule, which would minimize the operational overhead of rotating the credentials on a monthly basis.

upvoted 3 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option B, storing the credentials as secrets in AWS Systems Manager and using multi-Region secret replication, would not provide automatic rotation of secrets on a schedule.

Option C, storing the credentials in an S3 bucket with SSE enabled and using EventBridge to invoke an AWS Lambda function to rotate the credentials, would not provide automatic rotation of secrets on a schedule.

Option D, encrypting the credentials as secrets using KMS multi-Region customer managed keys and storing the secrets in a DynamoDB global table, would not provide automatic rotation of secrets on a schedule and would require additional operational overhead to retrieve the secrets from DynamoDB and use the RDS API to rotate the secrets.

overhead to retrieve the secrets from DynamoDB and use the RDS API to rotate the secrets.

upvoted 3 times

Musti35 8 months, 3 weeks ago

Selected Answer: A

With Secrets Manager, you can store, retrieve, manage, and rotate your secrets, including database credentials, API keys, and other secrets. When you create a secret using Secrets Manager, it's created and managed in a Region of your choosing. Although scoping secrets to a Region is a security best practice, there are scenarios such as disaster recovery and cross-Regional redundancy that require replication of secrets across Regions. Secrets Manager now makes it possible for you to easily replicate your secrets to one or more Regions to support these scenarios.

upvoted 3 times

linux_admin 8 months, 3 weeks ago

Selected Answer: A

A. Store the credentials as secrets in AWS Secrets Manager. Use multi-Region secret replication for the required Regions. Configure Secrets Manager to rotate the secrets on a schedule.

This solution is the best option for meeting the requirements with the least operational overhead. AWS Secrets Manager is designed specifically for managing and rotating secrets like database credentials. Using multi-Region secret replication, you can easily replicate the secrets across the required AWS Regions. Additionally, Secrets Manager allows you to configure automatic secret rotation on a schedule, further reducing the operational overhead.

upvoted 1 times

PaulGa 10 months ago

Selected Answer: A

Ans A. What's to debate...?

upvoted 1 times

creamymangosauce 11 months ago

Selected Answer: A

A - Secrets Manager automates the rotation of secrets for RDS without own implementation required, the options require effort to implement the secret rotation logic

upvoted 1 times

ics_911 1 year, 4 months ago

Selected Answer: A

A is correct.

upvoted 1 times

A_jaa 1 year, 5 months ago

Selected Answer: A

Answer-A

upvoted 1 times

gldiazcardenas 1 year, 8 months ago

Selected Answer: A

Clearly A is the correct one.

upvoted 1 times

TariqKipkemei 1 year, 10 months ago

Selected Answer: A

'The company needs to rotate the credentials for its Amazon RDS for MySQL databases across multiple AWS Regions' = AWS Secrets Manager

upvoted 1 times

miki111 1 year, 11 months ago

Option A MET THE REQUIREMENT

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #14 *Topic 1*

A company runs an ecommerce application on Amazon EC2 instances behind an Application Load Balancer. The instances run in an Amazon EC2 Auto Scaling group across multiple Availability Zones. The Auto Scaling group scales based on CPU utilization metrics. The ecommerce application stores the transaction data in a MySQL 8.0 database that is hosted on a large EC2 instance.
The database's performance degrades quickly as application load increases. The application handles more read requests than write transactions. The company wants a solution that will automatically scale the database to meet the demand of unpredictable read workloads while maintaining high availability.
Which solution will meet these requirements?

A. Use Amazon Redshift with a single node for leader and compute functionality.

B. Use Amazon RDS with a Single-AZ deployment Configure Amazon RDS to add reader instances in a different Availability Zone.

C. Use Amazon Aurora with a Multi-AZ deployment. Configure Aurora Auto Scaling with Aurora Replicas. **Most Voted**

D. Use Amazon ElastiCache for Memcached with EC2 Spot Instances.

Correct Answer: *C*

Community vote distribution

C (100%)

Comments

D2w Highly Voted 2 years, 8 months ago

Selected Answer: C

C, AURORA is 5x performance improvement over MySQL on RDS and handles more read requests than write,; maintaining high availability = Multi-AZ deployment

upvoted 51 times

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: C

Option C, using Amazon Aurora with a Multi-AZ deployment and configuring Aurora Auto Scaling with Aurora Replicas, would be the best solution to meet the requirements.

Aurora is a fully managed, MySQL-compatible relational database that is designed for high performance and high availability.

Aurora Multi-AZ deployments automatically maintain a synchronous standby replica in a different Availability Zone to provide high availability. Additionally, Aurora Auto Scaling allows you to automatically scale the number of Aurora Replicas in response to read workloads, allowing you to meet the demand of unpredictable read workloads while maintaining high availability. This would provide an automated solution for scaling the database to meet the demand of the application while maintaining high availability.

upvoted 25 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option A, using Amazon Redshift with a single node for leader and compute functionality, would not provide high availability.

Option B, using Amazon RDS with a Single-AZ deployment and configuring RDS to add reader instances in a different Availability Zone, would not provide high availability and would not automatically scale the number of reader instances in response to read workloads.

Option D, using Amazon ElastiCache for Memcached with EC2 Spot Instances, would not provide a database solution and would not meet the requirements.

upvoted 9 times

melvis8 Most Recent 2 months, 3 weeks ago

Selected Answer: C

Using aurora is the best answer for this issue since aurora is ideal for upredictable workloads and we can manage read heavy workloads by deploying the database with aurora replicas

upvoted 2 times

Tjazz04 6 months ago

Selected Answer: C

Keyword: High Availability
Only letter C has multi-AZ deployment. Also, it automatically scales replicas for unpredictable read workload

upvoted 2 times

VINVIN99 7 months, 3 weeks ago

Selected Answer: C

CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC

upvoted 1 times

PaulGa 10 months ago

Selected Answer: C

Ans C. Its the only one that offers right level of scaling and availability

upvoted 2 times

OBIOHAnze 1 year ago

Selected Answer: C

Here's why C is the best solution:

Amazon Aurora: A managed, high-performance MySQL-compatible relational database engine.
Multi-AZ deployment: Ensures high availability in case of an AZ failure.
Aurora Auto Scaling with Aurora Replicas: Automatically scales read replicas based on traffic, improving read performance.

upvoted 3 times

A_jaa 1 year, 5 months ago

Selected Answer: C

Answer-c

upvoted 1 times

Ndlesty 1 year, 7 months ago

Selected Answer: C

key statement: "...will automatically scale the database to meet the demand of unpredictable read workloads while maintaining high availability.

upvoted 2 times

AWSGuru123 1 year, 8 months ago

Selected Answer: C

Aurora
upvoted 1 times

Syruis 1 year, 10 months ago

Selected Answer: C

C fit perfectly
upvoted 1 times

TariqKipkemei 1 year, 10 months ago

Selected Answer: C

Unpredictable read workloads while maintaining high availability = Amazon Aurora with a Multi-AZ deployment, Auto Scaling with Aurora read replicas.
upvoted 1 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: C

As the application handles more read requests than write transactions, using read replicas with Aurora is an ideal choice as it allows read scaling without sacrificing write performance on the primary instance.
upvoted 2 times

miki111 1 year, 11 months ago

Option C MET THE REQUIREMENT
upvoted 1 times

hiepdz98 1 year, 11 months ago

Selected Answer: C

Option C
upvoted 1 times

cookieMr 1 year, 12 months ago

Selected Answer: C

Option C: Using Amazon Aurora with a Multi-AZ deployment and configuring Aurora Auto Scaling with Aurora Replicas is the most appropriate solution. Aurora is a MySQL-compatible relational database engine that provides high performance and scalability. With Multi-AZ deployment, the database is automatically replicated across multiple Availability Zones for high availability. Aurora Auto Scaling allows the database to automatically add or remove Aurora Replicas based on the workload, ensuring that read requests can be distributed effectively and the database can scale to meet demand. This provides both high availability and automatic scaling to handle unpredictable read workloads.
upvoted 2 times

Bmarodi 2 years ago

Selected Answer: C

C meets the requirements.
upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #15 *Topic 1*

A company recently migrated to AWS and wants to implement a solution to protect the traffic that flows in and out of the production VPC. The company had an inspection server in its on-premises data center. The inspection server performed specific operations such as traffic flow inspection and traffic filtering. The company wants to have the same functionalities in the AWS Cloud.
Which solution will meet these requirements?

A. Use Amazon GuardDuty for traffic inspection and traffic filtering in the production VPC.

B. Use Traffic Mirroring to mirror traffic from the production VPC for traffic inspection and filtering.

C. Use AWS Network Firewall to create the required rules for traffic inspection and traffic filtering for the production VPC.
Most Voted

D. Use AWS Firewall Manager to create the required rules for traffic inspection and traffic filtering for the production VPC.

Correct Answer: *C*

Community vote distribution

C (95%) B (5%)

Comments

SilentMilli Highly Voted 8 months, 3 weeks ago

Selected Answer: C

I would recommend option C: Use AWS Network Firewall to create the required rules for traffic inspection and traffic filtering for the production VPC.

AWS Network Firewall is a managed firewall service that provides filtering for both inbound and outbound network traffic. It allows you to create rules for traffic inspection and filtering, which can help protect your production VPC.

Option A: Amazon GuardDuty is a threat detection service, not a traffic inspection or filtering service.

Option B: Traffic Mirroring is a feature that allows you to replicate and send a copy of network traffic from a VPC to another VPC or on-premises location. It is not a service that performs traffic inspection or filtering.

Option D: AWS Firewall Manager is a security management service that helps you to centrally configure and manage firewalls across your accounts. It is not a service that performs traffic inspection or filtering.

upvoted 144 times

BoboChow Highly Voted 8 months, 3 weeks ago

Selected Answer: C

I agree with C.
AWS Network Firewall is a stateful, managed network firewall and intrusion detection and prevention service for your virtual private cloud (VPC) that you created in Amazon Virtual Private Cloud (Amazon VPC). With Network Firewall, you can filter traffic at the perimeter of your VPC. This includes filtering traffic going to and coming from an internet gateway, NAT gateway, or over VPN or AWS Direct Connect.

upvoted 26 times

BoboChow 2 years, 8 months ago

And I'm not sure Traffic Mirroring can be for filtering

upvoted 3 times

Juju66 Most Recent 1 month, 3 weeks ago

Selected Answer: C

I agree C.

Option D AWS Firewall Manager sounds similar, but it's not a service that can do traiffic inspection.

upvoted 1 times

satyaammm 5 months, 2 weeks ago

Selected Answer: C

AWS Firewall Manager is the only one that actually helps direct the network traffic in AWS. GuardDuty and AWS Firewall Manager recognized inspection against threats.

upvoted 1 times

Tjazz04 6 months ago

Selected Answer: C

C is the most appropriate when it comes to traffic flow inspection and filtering

upvoted 1 times

cookieMr 8 months, 3 weeks ago

Selected Answer: C

AWS Network Firewall is a managed network firewall service that allows you to define firewall rules to filter and inspect network traffic. You can create rules to define the traffic that should be allowed or blocked based on various criteria such as source/destination IP addresses, protocols, ports, and more. With AWS Network Firewall, you can implement traffic inspection and filtering capabilities within the production VPC, helping to protect the network traffic.

In the context of the given scenario, AWS Network Firewall can be a suitable choice if the company wants to implement traffic inspection and filtering directly within the VPC without the need for traffic mirroring. It provides an additional layer of security by enforcing specific rules for traffic filtering, which can help protect the production environment.

upvoted 3 times

Guru4Cloud 8 months, 3 weeks ago

Selected Answer: C

- AWS Network Firewall is a managed network security service that provides stateful inspection of traffic and allows you to define firewall rules to control the traffic flow in and out of your VPC.
- With AWS Network Firewall, you can create custom rule groups to define specific operations for traffic inspection and filtering.
- It can perform deep packet inspection and filtering at the network level to enforce security policies, block malicious traffic, and allow or deny traffic based on defined rules.
- By integrating AWS Network Firewall with the production VPC, you can achieve similar functionalities as the on-premises inspection server, performing traffic flow inspection and filtering.

upvoted 2 times

PaulGa 10 months ago

Selected Answer: C

Ans C. As per good response by SilentMili

upvoted 1 times

TheFivePips 1 year, 4 months ago

Selected Answer: C

I didn't realize the network firewall could do inspection, but here's what the documentation says:

AWS Network Firewall supports Transport Layer Security (TLS) inspection, allowing customers to strengthen their security posture on AWS by improving visibility into encrypted traffic flows. You can use AWS Network Firewall to decrypt TLS sessions and inspect both inbound and outbound Amazon Virtual Private Cloud (VPC) traffic without the need to deploy or manage any additional network security infrastructure. Encryption and decryption happen on the same firewall instance natively, so traffic does not cross any network boundaries.

upvoted 4 times

awsgeek75 1 year, 5 months ago

Selected Answer: C

Network Firewall to define firewall rules for traffic inspection.
A: GuardDuty is not for this
B: Wrong product
D: Firewall Manager does not monitor traffic, it manages firewall

upvoted 1 times

A_jaa 1 year, 5 months ago

Selected Answer: C

Answer-C

upvoted 1 times

danielpark99 1 year, 8 months ago

Selected Answer: C

AWS Nework Firewall to support from layert 3 to layer 7 protection, it is able to inspect any direction lets say vpc to vpc and outbound and inbound and even supporting direct connect and site to site vpn

upvoted 1 times

reema908516 1 year, 9 months ago

Selected Answer: C

AWS Network Firewall is a managed firewall service that provides filtering for both inbound and outbound network traffic. It allows you to create rules for traffic inspection and filtering, which can help protect your production VPC.

upvoted 1 times

nmywrld 1 year, 9 months ago

Why isn't D viable? Firewall Manager will help to provision network firewall as required if you define it in firewall manager. And it's fully managed, not requiring you to do any configuration or set up.

upvoted 1 times

pentium75 1 year, 5 months ago

Because we need a firewall, not a service that we COULD IN THEORY use to create a firewall?

upvoted 2 times

Syruis 1 year, 10 months ago

Selected Answer: C

C with no doubt

upvoted 1 times

miki111 1 year, 11 months ago

Option C MET THE REQUIREMENT

upvoted 1 times

AJAYSINGH0807 2 years ago

B is correct answer

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #16 *Topic 1*

A company hosts a data lake on AWS. The data lake consists of data in Amazon S3 and Amazon RDS for PostgreSQL. The company needs a reporting solution that provides data visualization and includes all the data sources within the data lake. Only the company's management team should have full access to all the visualizations. The rest of the company should have only limited access.
Which solution will meet these requirements?

A. Create an analysis in Amazon QuickSight. Connect all the data sources and create new datasets. Publish dashboards to visualize the data. Share the dashboards with the appropriate IAM roles.

B. Create an analysis in Amazon QuickSight. Connect all the data sources and create new datasets. Publish dashboards to visualize the data. Share the dashboards with the appropriate users and groups. **Most Voted**

C. Create an AWS Glue table and crawler for the data in Amazon S3. Create an AWS Glue extract, transform, and load (ETL) job to produce reports. Publish the reports to Amazon S3. Use S3 bucket policies to limit access to the reports.

D. Create an AWS Glue table and crawler for the data in Amazon S3. Use Amazon Athena Federated Query to access data within Amazon RDS for PostgreSQL. Generate reports by using Amazon Athena. Publish the reports to Amazon S3. Use S3 bucket policies to limit access to the reports.

Correct Answer: *B*

Community vote distribution

B (86%) Other (14%)

Comments

rodriiviru Highly Voted 2 years, 8 months ago

Selected Answer: B

B

upvoted 71 times

BoboChow 2 years, 8 months ago

Agree with you

upvoted 3 times

PhucVuu Highly Voted 8 months, 3 weeks ago

Selected Answer: B

Keywords:
- Data lake on AWS.
- Consists of data in Amazon S3 and Amazon RDS for PostgreSQL.
- The company needs a reporting solution that provides data VISUALIZATION and includes ALL the data sources within the data lake.

A - Incorrect: Amazon QuickSight only support users(standard version) and groups (enterprise version). users and groups only exists without QuickSight. QuickSight don't support IAM. We use users and groups to view the QuickSight dashboard
B - Correct: as explained in answer A and QuickSight is used to created dashboard from S3, RDS, Redshift, Aurora, Athena, OpenSearch, Timestream
C - Incorrect: This way don't support visulization and don't mention how to process RDS data
D - Incorrect: This way don't support visulization and don't mention how to combine data RDS and S3
upvoted 60 times

Sanki_26 Most Recent 2 months, 1 week ago

Selected Answer: B

Amazon QuickSight is a data visualization service that allows you to create interactive dashboards and reports from various data sources, including Amazon S3 and Amazon RDS for PostgreSQL. You can connect all the data sources and create new datasets in QuickSight, and then publish dashboards to visualize the data. You can also share the dashboards with the appropriate users and groups, and control their access levels using IAM roles and permissions.
upvoted 2 times

Vandaman 3 months, 3 weeks ago

Selected Answer: B

It was the use of users and groups that answered that for me
upvoted 1 times

FlyingHawk 5 months ago

Selected Answer: B

Amazon QuickSight can handle dynamic data through Direct Query Mode or scheduled data refreshes. It provides real-time or near-real-time visualizations, interactive dashboards, and fine-grained access control, making it the most suitable solution for the company's requirements.

https://aws.amazon.com/blogs/business-intelligence/best-practices-for-amazon-quicksight-spice-and-direct-query-mode/

I saw the question in udemy practice test 5 from Neal Davis, he suggested the correct answer is D, his reasons: option B solves the problem of access sharing with resources but does not take care of delta in data.
upvoted 2 times

MGKYAING 5 months, 2 weeks ago

Selected Answer: B

Amazon QuickSight is designed for creating visualizations and dashboards, and it can easily connect to a variety of data sources such as Amazon S3 and Amazon RDS for PostgreSQL.
You can create an analysis in QuickSight by connecting the required data sources, creating datasets, and then building dashboards.
Access control in QuickSight is robust. You can share dashboards with specific users and groups, ensuring that only the management team has full access to all the visualizations, while the rest of the company can have limited access.
upvoted 1 times

Tjazz04 6 months ago

Selected Answer: B

Keyword: data visualization = QuickSight
QuickSight defines Users (standard versions) and Groups (enterprise version)
These users & groups only exist within QuickSight, not IAM
So the correct answer is B.
upvoted 1 times

EzKkk 7 months ago

Selected Answer: B

The question requires a solution for data visualization, which means it focuses solely on downstream consumption. Therefore, any solution that includes upstream processing is off the table (C & D). This leaves us with two options: A and B. Essentially, they are the same with one key difference: A manages access through IAM Roles, while B manages it through an IAM group they are the same with one key difference: A manages access through IAM Roles, while B manages it through an IAM group. Since only the management team is granted permission, using a group will be more efficient, as it scales better with changes in personnel and provides a centralized point for managing permissions.

upvoted 1 times

EzKkk 7 months ago

The question requires a solution for data visualization, which means it focuses solely on downstream consumption. Therefore, any solution that includes upstream processing is off the table (C & D). This leaves us with two options: A and B. Essentially, they are the same with one key difference: A manages access through IAM Roles, while B manages it through an IAM group. Since only the management team is granted permission, using a group will be more efficient, as it scales better with changes in personnel and provides a centralized point for managing permissions.

upvoted 1 times

hanen 8 months, 3 weeks ago

Selected Answer: D

If you have data in sources other than Amazon S3, you can use Athena Federated Query to query the data in place or build pipelines that extract data from multiple data sources and store them in Amazon S3. With Athena Federated Query, you can run SQL queries across data stored in relational, non-relational, object, and custom data sources.

Athena uses data source connectors that run on AWS Lambda to run federated queries. A data source connector is a piece of code that can translate between your target data source and Athena. You can think of a connector as an extension of Athena's query engine. Prebuilt Athena data source connectors exist for data sources like Amazon CloudWatch Logs, Amazon DynamoDB, Amazon DocumentDB, and Amazon RDS, and JDBC-compliant relational data sources such MySQL, and PostgreSQL under the Apache 2.0 license

upvoted 4 times

FlyingHawk 5 months ago

D relies on static reports stored in Amazon S3.
If the data changes, the reports would need to be regenerated and re-published, which is not ideal for dynamic data.
They also lack interactive visualization capabilities and fine-grained access control for dashboards.

upvoted 1 times

JA2018 7 months, 1 week ago

key phases: " The data lake consists of data in Amazon S3 and Amazon RDS for PostgreSQL. The company needs a reporting solution that provides data visualization and includes all the data sources within the data lake. "

upvoted 1 times

linux_admin 8 months, 3 weeks ago

Selected Answer: B

Option B is the correct answer because Amazon QuickSight's sharing mechanism is based on users and groups, not IAM roles. IAM roles are used for granting permissions to AWS resources, but they are not directly used for sharing QuickSight dashboards.

In option B, you create an analysis in Amazon QuickSight, connect all the data sources (Amazon S3 and Amazon RDS for PostgreSQL), and create new datasets. After publishing dashboards to visualize the data, you share them with appropriate users and groups. This approach allows you to control the access levels for different users, such as providing full access to the management team and limited access to the rest of the company. This solution meets the requirements specified in the question.

upvoted 7 times

elearningtakai 8 months, 3 weeks ago

Selected Answer: B

Amazon QuickSight is a cloud-based business intelligence (BI) service that makes it easy to create and publish interactive dashboards that include data visualizations from multiple data sources. By using QuickSight, the company can connect to both Amazon S3 and Amazon RDS for PostgreSQL and create new datasets that combine data from both sources. The company can then use QuickSight to create interactive dashboards that visualize the data and provide data insights.

To limit access to the visualizations, the company can use QuickSight's built-in security features. QuickSight allows you to define fine-grained access control at the user or group level. This way, the management team can have full access to all the visualizations, while the rest of the company can have only limited access.

upvoted 4 times

dszes 8 months, 3 weeks ago

tricky question, Users, groups and roles can have access.
Viewing who has access to a dashboard

Viewing who has access to a dashboard
Use the following procedure to see which users or groups have access to the dashboard.

Open the published dashboard and choose Share at upper right. Then choose Share dashboard.

In the Share dashboard page that opens, under Manage permissions, review the users and groups, and their roles and settings.

You can search to locate a specific user or group by entering their name or any part of their name in the search box at upper right. Searching is case-sensitive, and wildcards aren't supported. Delete the search term to return the view to all users.
upvoted 2 times

cookieMr 8 months, 3 weeks ago

Selected Answer: B

B. Create an analysis in Amazon QuickSight. Connect all the data sources and create new datasets. Publish dashboards to visualize the data. Share the dashboards with the appropriate users and groups.

Amazon QuickSight is a business intelligence (BI) tool provided by AWS that allows you to create interactive dashboards and reports. It supports a variety of data sources, including Amazon S3 and Amazon RDS for PostgreSQL, which are the data sources in the company's data lake.

Option A (Create an analysis in Amazon QuickSight and share with IAM roles) is incorrect because it suggests sharing with IAM roles, which are more suitable for managing access to AWS resources rather than granting access to specific users or groups within QuickSight.
upvoted 4 times

Guru4Cloud 8 months, 3 weeks ago

Selected Answer: B

Explanation:

Option B involves using Amazon QuickSight, which is a business intelligence tool provided by AWS for data visualization and reporting. With this option, you can connect all the data sources within the data lake, including Amazon S3 and Amazon RDS for PostgreSQL. You can create datasets within QuickSight that pull data from these sources.

The solution allows you to publish dashboards in Amazon QuickSight, which will provide the required data visualization capabilities. To control access, you can use appropriate IAM (Identity and Access Management) roles, assigning full access only to the company's management team and limiting access for the rest of the company. You can share the dashboards selectively with the users and groups that need access.
upvoted 3 times

IdanAWS 8 months, 3 weeks ago

My opinion is divided here, and I will explain:
Option C can be correct because glue crawler is used to access S3, and athena federated query is used to access RDS.
My problem with answer C is that it says:
"Generate Reports by using athena"
And I think that is not true. athena alone does not generate reports, it has to integrate with services such as quickSight and then it generates reports, therefore the answer is not written properly and I think C is a mistake.
Since C is wrong I think B is the correct answer.
upvoted 2 times

Ruffyit 8 months, 3 weeks ago

Explanation:

Option B involves using Amazon QuickSight, which is a business intelligence tool provided by AWS for data visualization and reporting. With this option, you can connect all the data sources within the data lake, including Amazon S3 and Amazon RDS for PostgreSQL. You can create datasets within QuickSight that pull data from these sources.

The solution allows you to publish dashboards in Amazon QuickSight, which will provide the required data visualization capabilities. To control access, you can use appropriate IAM (Identity and Access Management) roles, assigning full access only to the company's management team and limiting access for the rest of the company. You can share the dashboards selectively with the users and groups that need access.
upvoted 1 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #17 *Topic 1*

A company is implementing a new business application. The application runs on two Amazon EC2 instances and uses an Amazon S3 bucket for document storage. A solutions architect needs to ensure that the EC2 instances can access the S3 bucket.
What should the solutions architect do to meet this requirement?

A. Create an IAM role that grants access to the S3 bucket. Attach the role to the EC2 instances. **Most Voted**

B. Create an IAM policy that grants access to the S3 bucket. Attach the policy to the EC2 instances.

C. Create an IAM group that grants access to the S3 bucket. Attach the group to the EC2 instances.

D. Create an IAM user that grants access to the S3 bucket. Attach the user account to the EC2 instances.

Correct Answer: *A*

Community vote distribution

A (100%)

Comments

sba21 Highly Voted 2 years, 8 months ago

Selected Answer: A

Always remember that you should associate IAM roles to EC2 instances

upvoted 103 times

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: A

The correct option to meet this requirement is A: Create an IAM role that grants access to the S3 bucket and attach the role to the EC2 instances.

An IAM role is an AWS resource that allows you to delegate access to AWS resources and services. You can create an IAM role that grants access to the S3 bucket and then attach the role to the EC2 instances. This will allow the EC2 instances to access the S3 bucket and the documents stored within it.

Option B is incorrect because an IAM policy is used to define permissions for an IAM user or group, not for an EC2 instance.

Option C is incorrect because an IAM group is used to group together IAM users and policies, not to grant access to resources.

Option D is incorrect because an IAM user is used to represent a person or service that interacts with AWS resources, not to grant access to resources.

grant access to resources.

upvoted 74 times

satyaammm Most Recent 5 months, 2 weeks ago

Selected Answer: A

Only IAM role is suitable here as policies are added to IAM groups or users. Moreover, IAM groups cannot be attached to a resource and attaching IAM user credentials to a IAM role in risky and thereby not suitable.

upvoted 2 times

Tjazz04 6 months ago

Selected Answer: A

Ans. A

Here's why:

IAM Role: Roles are designed to be assumed by entities like EC2 instances. By creating an IAM role with the necessary permissions to access the S3 bucket and attaching this role to the EC2 instances, you ensure that the instances can securely access the S3 bucket without needing to manage long-term credentials.
IAM Policy: While policies define permissions, they need to be attached to roles or users. Attaching a policy directly to EC2 instances is not possible.
IAM Group: Groups are used to manage permissions for multiple users, not instances.
IAM User: Users are intended for individual people or applications, not for EC2 instances.
By using an IAM role, you follow AWS best practices for security and manageability. If you have any more questions or need further clarification, feel free to ask!

upvoted 2 times

EzKkk 7 months ago

Selected Answer: A

IAM Role + EC2 instance = go-to solution

upvoted 1 times

PaulGa 10 months ago

Selected Answer: A

Ans A - as per "Buruguduystunstugudunstuy" response.

upvoted 1 times

A_jaa 1 year, 5 months ago

Selected Answer: A

Answer-A

upvoted 2 times

thewalker 1 year, 5 months ago

Selected Answer: A

Below is the response from Amazon Q:

To access S3 from an EC2 instance, you need to create an IAM role and associate that role with the EC2 instance. Here are the basic steps:
1. Create an IAM role and attach the AmazonS3ReadOnlyAccess or AmazonS3FullAccess managed policy to grant S3 access.
2. Launch the EC2 instance and select the IAM role you created during launch.
3. The instance will now have the permissions defined in the IAM role and you can access S3 from the instance.

upvoted 2 times

thewalker 1 year, 5 months ago

Some key points:
1. Attaching an IAM role is preferred over creating a resource-based policy for S3 access from EC2 as it provides centralized access management.
2. The instance will need internet access to communicate with S3. Make sure the associated security group and NACL rules allow outbound internet access.
3. Check AWS documentation for latest steps to create and associate an IAM role with an EC2 instance. The console and CLI provide options to automate this process.

upvoted 1 times

jjcode 1 year, 6 months ago

Strangely straight forward, Almost had me confused.
upvoted 1 times

GabrielSGoncalves 1 year, 7 months ago

Selected Answer: A

For sure
upvoted 1 times

Ruffyit 1 year, 7 months ago

The correct option to meet this requirement is A: Create an IAM role that grants access to the S3 bucket and attach the role to the EC2 instances.

An IAM role is an AWS resource that allows you to delegate access to AWS resources and services. You can create an IAM role that grants access to the S3 bucket and then attach the role to the EC2 instances. This will allow the EC2 instances to access the S3 bucket and the documents stored within it.

Option B is incorrect because an IAM policy is used to define permissions for an IAM user or group, not for an EC2 instance.

Option C is incorrect because an IAM group is used to group together IAM users and policies, not to grant access to resources.

Option D is incorrect because an IAM user is used to represent a person or service that interacts with AWS resources, not to grant access to resources.
upvoted 1 times

danielpark99 1 year, 8 months ago

Selected Answer: A

EC2 instances should be associated with IAM roles.
Policies can be applying to users and groups can help to apply multiple roles.
upvoted 1 times

Abdou1604 1 year, 10 months ago

Option B may work but ,
suggests creating an IAM policy directly and attaching it to the EC2 instances. While this might work, it's not the recommended approach. Using an IAM role is more secure and manageable.
upvoted 1 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: A

Always remember that you should associate IAM roles to EC2 instances.
An IAM role is an AWS resource that allows you to delegate access to AWS resources and services. You can create an IAM role that grants access to the S3 bucket and then attach the role to the EC2 instances. This will allow the EC2 instances to access the S3 bucket and the documents stored within it.
upvoted 1 times

Rexino 1 year, 11 months ago

Selected Answer: A

IAM roles should be associated to EC2 instance
upvoted 2 times

miki111 1 year, 11 months ago

Option A MET THE REQUIREMENT
upvoted 1 times

cookieMr 1 year, 12 months ago

Selected Answer: A

Option A is the correct approach because IAM roles are designed to provide temporary credentials to AWS resources such as EC2 instances. By creating an IAM role, you can define the necessary permissions and policies that allow the EC2 instances to access the S3 bucket securely. Attaching the IAM role to the EC2 instances will automatically provide the necessary credentials to access the S3 bucket without the need for explicit access keys or secrets.

Option B is not recommended in this case because IAM policies alone cannot be directly attached to EC2 instances. Policies are usually attached to IAM users, groups, or roles.

Option C is not the most appropriate choice because IAM groups are used to manage collections of IAM users and their permissions, rather than granting access to specific resources like S3 buckets.

Option D is not the optimal solution because IAM users are intended for individual user accounts and are not the recommended approach for granting access to resources within EC2 instances.

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #18 *Topic 1*

An application development team is designing a microservice that will convert large images to smaller, compressed images. When a user uploads an image through the web interface, the microservice should store the image in an Amazon S3 bucket, process and compress the image with an AWS Lambda function, and store the image in its compressed form in a different S3 bucket.
A solutions architect needs to design a solution that uses durable, stateless components to process the images automatically.
Which combination of actions will meet these requirements? (Choose two.)

A. Create an Amazon Simple Queue Service (Amazon SQS) queue. Configure the S3 bucket to send a notification to the SQS queue when an image is uploaded to the S3 bucket. Most Voted

B. Configure the Lambda function to use the Amazon Simple Queue Service (Amazon SQS) queue as the invocation source. When the SQS message is successfully processed, delete the message in the queue. Most Voted

C. Configure the Lambda function to monitor the S3 bucket for new uploads. When an uploaded image is detected, write the file name to a text file in memory and use the text file to keep track of the images that were processed.

D. Launch an Amazon EC2 instance to monitor an Amazon Simple Queue Service (Amazon SQS) queue. When items are added to the queue, log the file name in a text file on the EC2 instance and invoke the Lambda function.

E. Configure an Amazon EventBridge (Amazon CloudWatch Events) event to monitor the S3 bucket. When an image is uploaded, send an alert to an Amazon ample Notification Service (Amazon SNS) topic with the application owner's email address for further processing.

Correct Answer: *AB*

Community vote distribution

AB (99%) A

Comments

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: AB

To design a solution that uses durable, stateless components to process images automatically, a solutions architect could consider the following actions:

Option A involves creating an SQS queue and configuring the S3 bucket to send a notification to the queue when an image is uploaded. This allows the application to decouple the image upload process from the image processing process and ensures that the image processing process is triggered automatically when a new image is uploaded.

the image processing process is triggered automatically when a new image is uploaded.

Option B involves configuring the Lambda function to use the SQS queue as the invocation source. When the SQS message is successfully processed, the message is deleted from the queue. This ensures that the Lambda function is invoked only once per image and that the image is not processed multiple times.

upvoted 43 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option C is incorrect because it involves storing state (the file name) in memory, which is not a durable or scalable solution.

Option D is incorrect because it involves launching an EC2 instance to monitor the SQS queue, which is not a stateless solution.

Option E is incorrect because it involves using Amazon EventBridge (formerly Amazon CloudWatch Events) to send an alert to an Amazon Simple Notification Service (Amazon SNS) topic, which is not related to the image processing process.

upvoted 24 times

hsinchang 1 year, 10 months ago

So storing states invokes the stateless principle, nice understanding!

upvoted 2 times

op22233 1 year, 8 months ago

A stateless system sends a request to the server and relays the response (or the state) back without storing any information. On the other hand, stateful systems expect a response, track information, and resend the request if no response is received

upvoted 4 times

sba21 Highly Voted 2 years, 8 months ago

Selected Answer: AB

It looks like A-B

upvoted 15 times

vadps Most Recent 2 weeks ago

Selected Answer: AB

AB: S3-SQS-Lambda

upvoted 1 times

EzKkk 7 months ago

Selected Answer: AB

High level flow: Web (1) -> Internet (2) -> Uploaded image bucket (3) -> Compressed image bucket (4)
Let's break this down.
Image uploading (1)(2)(3): If the image's size is big, multi upload is a MUST. If you want to accelerate the process, you can also use Acceleration Transfer with additional fee.
Image compressing (3)(4): SQS Simple Queue can directly integrate with S3 events. For every image uploaded, S3 event will create a message in SQS queue. After that, Lambda can be invoked to compress the image and then upload the image to destination bucket.

upvoted 2 times

PhucVuu 8 months, 3 weeks ago

Selected Answer: AB

Keywords:
- Store the image in an Amazon S3 bucket, process and compress the image with an AWS Lambda function.
- Durable, stateless components to process the images automatically

A,B: Correct - SQS has message retention function(store message) default 4 days(can increate update 14 days) so that you can re-run lambda if there are any errors when processing the images.
C: Incorrect - Lambda function just run the request then stop, the max tmeout is 15 mins. So we cannot store data in the ram of Lambda function.
D: Incorrect - we can trigger Lambda dirrectly from SQS no need EC2 instance in this case
E: Incorrect - It kinds of manually step -> the owner has to read email then process it :))

upvoted 9 times

Guru4Cloud 8 months, 3 weeks ago

Selected Answer: AB

Explanation:

Option A: By creating an Amazon SQS queue and configuring the S3 bucket to send a notification to the SQS queue when an image is uploaded, the system establishes a durable and scalable way to handle incoming image processing tasks.

Option B: Configuring the Lambda function to use the SQS queue as the invocation source allows it to retrieve messages from the queue and process them in a stateless manner. After successfully processing the image, the Lambda function can delete the message from the queue to avoid duplicate processing.

upvoted 3 times

DigitalDanny 8 months, 3 weeks ago

Selected Answer: AB

A. Create an Amazon Simple Queue Service (Amazon SQS) queue. Configure the S3 bucket to send a notification to the SQS queue when an image is uploaded to the S3 bucket.

B. Configure the Lambda function to use the Amazon Simple Queue Service (Amazon SQS) queue as the invocation source. When the SQS message is successfully processed, delete the message in the queue.

Explanation:

A (SQS Queue): Using SQS to decouple the S3 bucket from the processing components provides durability and scalability. When an image is uploaded, a notification is sent to the SQS queue.
B (Lambda with SQS Trigger): Configuring the Lambda function to use the SQS queue as the invocation source allows for stateless and scalable image processing. Lambda can be triggered by messages in the SQS queue, and upon successful processing, the message can be deleted, ensuring that each message (image) is processed once.
This combination ensures a durable, stateless, and scalable architecture for processing images automatically in response to user uploads.

upvoted 3 times

cookieMr 8 months, 3 weeks ago

Selected Answer: AB

Option A is a correct because it allows for decoupling between the image upload process and image processing. By configuring S3 to send a notification to SQS, image upload event is recorded and can be processed independently by microservice.

Option B is also a correct because it ensures that Lambda is triggered by messages in SQS. Lambda can retrieve image information from SQS, process and compress image, and store compressed image in a different S3. Once processing is successful, Lambda can delete processed message from SQS, indicating that image has been processed.

Option C is not recommended because it introduces a stateful approach by using a text file to keep track of processed images.

Option D is not optimal solution as it introduces unnecessary complexity by involving an EC2 to monitor SQS and maintain a text file.

Option E is not directly related to requirement of processing images automatically. Although EventBridge and SNS can be useful for event notifications and further processing, they don't provide the same level of durability and scalability as SQS.

upvoted 6 times

PaulGa 10 months ago

Selected Answer: AB

Ans A,B - as per Buruguduystunstugudunstuy's response: stateless, robust

upvoted 1 times

soufiyane 1 year, 2 months ago

Selected Answer: AB

whenever we talk about microservices we should mention SQS, so A and B are the right answers

upvoted 4 times

han_ds 1 year, 3 months ago

Selected Answer: AB

A/B make the most sense and in practice this works, I've done it.

upvoted 3 times

A_jaa 1 year, 5 months ago

Selected Answer: AB

Answer- A,B

upvoted 1 times

mohamedsambo 1 year, 5 months ago

I can not understand why it is not as simple like s3-1 event destination to notify the lambda function to process and upload to s3-2

upvoted 2 times

miki111 1 year, 11 months ago

Option AB MET THE REQUIREMENT

upvoted 1 times

RupeC 1 year, 11 months ago

Selected Answer: AB

D and E are distractions. C seems a valid solution. However, as you have to select two, A and B are the only two that work in conjunction with each other.

upvoted 2 times

tester0071 1 year, 11 months ago

Selected Answer: AB

A and B are optimal solutions

upvoted 1 times

beginnercloud 2 years ago

Selected Answer: AB

Option A nad B

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #19 *Topic 1*

A company has a three-tier web application that is deployed on AWS. The web servers are deployed in a public subnet in a VPC. The application servers and database servers are deployed in private subnets in the same VPC. The company has deployed a third-party virtual firewall appliance from AWS Marketplace in an inspection VPC. The appliance is configured with an IP interface that can accept IP packets.
A solutions architect needs to integrate the web application with the appliance to inspect all traffic to the application before the traffic reaches the web server.
Which solution will meet these requirements with the LEAST operational overhead?

A. Create a Network Load Balancer in the public subnet of the application's VPC to route the traffic to the appliance for packet inspection.

B. Create an Application Load Balancer in the public subnet of the application's VPC to route the traffic to the appliance for packet inspection.

C. Deploy a transit gateway in the inspection VPConfigure route tables to route the incoming packets through the transit gateway.

D. Deploy a Gateway Load Balancer in the inspection VPC. Create a Gateway Load Balancer endpoint to receive the incoming packets and forward the packets to the appliance. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (83%) Other (17%)

Comments

CloudGuru99 Highly Voted 2 years, 8 months ago

Answer is D . Use Gateway Load balancer

upvoted 49 times

pm2229 Highly Voted 8 months, 3 weeks ago

It's D, Coz.. Gateway Load Balancer is a new type of load balancer that operates at layer 3 of the OSI model and is built on Hyperplane, which is capable of handling several thousands of connections per second. Gateway Load Balancer endpoints are configured in spoke VPCs originating or receiving traffic from the Internet. This architecture allows you to perform inline inspection of traffic from multiple spoke VPCs in a simplified and scalable fashion while still centralizing your virtual appliances.

upvoted 47 times

surajkrishnamurthy Most Recent 4 months ago

Selected Answer: D

3rd Party Virtual Appliances ---> Gateway Loadbalancer

upvoted 4 times

EllenLiu 6 months ago

Selected Answer: D

we're able to load balance some of the incoming traffic through to those virtual appliances where they can perform some kind of inspection.

- Used in front of virtual appliances such as:
- firewalls,
- IDS: intrusion detection systems
- IPS: intrusion prevention systems
- DPI: deep packet inspection systems
- Operates at Layer 3 – listens for all packets on all ports

- Forwards traffic to the TG specified in the listener rules

- GLB and virtual appliances Exchanges traffic using the GENEVE protocol on port 6081

upvoted 1 times

carlton_agesa 5 months ago

In this context, "using the GENEVE protocol" refers to the method by which the Global Load Balancer (GLB) and the virtual appliances (like firewalls and intrusion detection systems) communicate and exchange traffic. GENEVE (Generic Network Virtualization Encapsulation) is a protocol that encapsulates packets for efficient transport over a network, allowing for flexible and scalable network virtualization. It operates on port 6081, enabling the load balancer to forward traffic to the specified target groups (TG) based on the listener rules.
Credits: AI answer

upvoted 2 times

EzKkk 7 months ago

Selected Answer: D

The answer is D for some obvious reasons:
- Low operational overhead.
- Integrate AWS Marketplace and third-party application.

I think the question will be much more interesting if the author add another option like using VPC-to-VPC traffic inspection because the question asked traffic to be inspected before it reaches application layer so traffic forwarding is also a feasible solution.

upvoted 1 times

Buruguduystunstugudunstuy 8 months, 3 weeks ago

Selected Answer: D

The solution that will meet these requirements with the least operational overhead is D: Deploy a Gateway Load Balancer in the inspection VPC and create a Gateway Load Balancer endpoint to receive the incoming packets and forward the packets to the appliance.

A Gateway Load Balancer is a fully managed service that provides a single point of contact for clients and distributes incoming traffic across multiple targets, such as Amazon Elastic Compute Cloud (EC2) instances and containers, in one or more virtual private clouds (VPCs). You can deploy a Gateway Load Balancer in the inspection VPC and create a Gateway Load Balancer endpoint to receive the incoming packets from the web servers in the application's VPC and forward the packets to the appliance for packet inspection. This will allow you to inspect all traffic to the web application with minimal operational overhead.

upvoted 8 times

JA2018 7 months, 1 week ago

Gateway Load Balancer helps you easily deploy, scale, and manage your third-party virtual appliances. It gives you one gateway for distributing traffic across multiple virtual appliances while scaling them up or down, based on demand. This decreases potential points of failure in your network and increases availability.

You can find, test, and buy virtual appliances from third-party vendors directly in AWS Marketplace. This integrated experience streamlines the deployment process so you see value from your virtual appliances more quickly—whether you want to keep working with your current vendors or try something new.

upvoted 1 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option A is incorrect because a Network Load Balancer is designed to handle traffic at the connection level and is not suitable for packet inspection.

Option B is incorrect because an Application Load Balancer is designed to handle traffic at the request level and is not suitable for packet inspection.

Option C is incorrect because a transit gateway is designed to allow multiple VPCs and on-premises networks to connect to each other, but it is not suitable for packet inspection.

upvoted 9 times

Bang3R 8 months, 3 weeks ago

D. Deploy a Gateway Load Balancer in the inspection VPC. Create a Gateway Load Balancer endpoint to receive the incoming packets and forward the packets to the appliance.

A Gateway Load Balancer can inspect traffic before forwarding it to a virtual appliance for additional processing. The solution will not require changing the existing architecture and will have the least amount of operational overhead. The appliance can be configured with a specific IP interface to accept IP packets. The Gateway Load Balancer can be configured with an endpoint to route incoming packets to the appliance. The solution ensures all traffic to the web application is inspected before it reaches the web server.

upvoted 3 times

PhucVuu 8 months, 3 weeks ago

Selected Answer: D

Keywords:Third-party virtual firewall appliance from AWS Marketplace in an inspection VPC -> only Gateway Load Balancer support it

A: Incorrect - Network Load Balancer don't support to route traffic to third-party virtual firewall appliance.
B: Incorrect - Application Load Balancer don't support to route traffic to third-party virtual firewall appliance.
C: Incorrect - Transit Gateway is use as connect center to connect all VPC, Direct Connect Gateway and VPN Connection. Routes Tables in Trasit Gateway only limit which VPC can talk to other VPCs.
D: Correct - Gateway Load Balancer support route traffic to third-party virtual firewall appliance in layer 3 that make it different from ALB and NLB.

upvoted 20 times

lht 8 months, 3 weeks ago

Selected Answer: B

In the scenario described, the web servers, application servers, and database servers are all located within the same VPC. Therefore, a Gateway Load Balancer may not be the most suitable choice for load balancing traffic between them.

Instead, an Application Load Balancer (ALB) would be a better option as it operates at Layer 7 and can inspect traffic at the application layer. This would allow the virtual firewall to inspect traffic before it reaches the web servers, which is the requirement specified in the scenario.

Overall, while a Gateway Load Balancer can be useful in certain scenarios, it is not the best choice for this particular use case. An Application Load Balancer is a better option as it provides the necessary features to integrate the web application with the virtual firewall appliance and inspect all traffic before it reaches the web server.

upvoted 2 times

johndoe 8 months, 3 weeks ago

Selected Answer: D

Here's why
Traffic enters the service consumer VPC through the internet gateway.
Traffic is sent to the Gateway Load Balancer endpoint, as a result of ingress routing.
Traffic is sent to the Gateway Load Balancer for inspection through the security appliance.
Traffic is sent back to the Gateway Load Balancer endpoint after inspection.
Traffic is sent to the application servers (destination subnet).
https://docs.aws.amazon.com/elasticloadbalancing/latest/gateway/getting-started.html
But I ain't completely sure about the least operational overhead.

upvoted 3 times

AlaTaftaf 8 months, 3 weeks ago

Selected Answer: B

This is the answer of ChatGpt:
Option B is the correct solution because the ALB can be used to redirect traffic to the virtual firewall appliance without requiring any changes to the backend application servers. The ALB can also be configured to send traffic to multiple targets, allowing the architect to perform high availability and load balancing. This solution is easy to implement and manage and does not require any additional components such as transit gateways or gateway load balancers.
Option D is not the optimal solution since Gateway Load Balancer (GWLB) is intended for use with virtual appliances in the cloud, such as firewalls and intrusion prevention systems. However, it adds operational overhead since creating and managing a Gateway Load Balancer requires several components, including an endpoint group and listener.

upvoted 2 times

cookieMr 8 months, 3 weeks ago

Selected Answer: A

A. Create a Network Load Balancer in the public subnet of the application's VPC to route the traffic to the appliance for packet inspection.

By creating a Network Load Balancer (NLB) in the public subnet, you can configure it to forward incoming traffic to the virtual firewall appliance for inspection. The NLB operates at the transport layer (Layer 4) and can distribute traffic across multiple instances, including the firewall appliance. This allows you to scale the inspection capacity if needed. The NLB can be associated with a target group that includes the IP address of the firewall appliance, directing traffic to it before reaching the web servers.

Option B (Application Load Balancer) is not suitable for this scenario as it operates at the application layer (Layer 7) and does not provide direct access to the IP packets for inspection.

Option C (Transit Gateway) and option D (Gateway Load Balancer) introduce additional complexity and overhead compared to using an NLB. They are not necessary for achieving the requirement of inspecting traffic to the web application before reaching the web servers.

upvoted 8 times

vipyodha 1 year, 11 months ago

best answer.well explained

upvoted 1 times

Guru4Cloud 8 months, 3 weeks ago

Selected Answer: D

The correct answer is D.
Here is the explanation:
Option D is correct because a Gateway Load Balancer (GWLB) is a global service, and it can be deployed in any VPC. This means that the GWLB can reach the appliance. Additionally, the GWLB can be configured to forward packets to the appliance for packet inspection.

Option A is incorrect because a Network Load Balancer (NLB) is a regional service, and the appliance is deployed in an inspection VPC. This means that the NLB would not be able to reach the appliance.
Option B is incorrect because an Application Load Balancer (ALB) is a regional service, and the appliance is deployed in an inspection VPC. This means that the ALB would not be able to reach the appliance.
Option C is incorrect because a transit gateway is a global service, and the appliance is deployed in an inspection VPC. This means that the transit gateway would not be able to reach the appliance.

upvoted 11 times

DigitalDanny 8 months, 3 weeks ago

Selected Answer: D

Gateway Load Balancer (GWLB): GWLB is designed for deploying third-party appliances and provides a scalable and easy way to route traffic through appliances. It operates at the network layer and can handle both TCP and UDP traffic.
Operational Overhead: Deploying a GWLB in the inspection VPC and creating an endpoint involves less operational overhead compared to managing Load Balancers in the application's VPC. It allows for centralized management of the inspection process.
This solution ensures that all traffic is routed through the Gateway Load Balancer for inspection before reaching the web servers, providing a scalable and efficient way to integrate the third-party virtual firewall appliance

upvoted 5 times

Cmtan 8 months, 3 weeks ago

A and B are wrong, as they don'
t support cross-VPC traffic routing

Option C -transit gateway attached to VPC,updating route table and configure security groups and network ACLs can accomplish the task.

Meanwhile, Gateway load balancer is designed meant for routing traffic across VPC, but itself alone does not work. All effort mentioned is C are still required. So this is not the least effort?

upvoted 1 times

PaulGa 10 months ago

Selected Answer: B

Ans B – (a) because it’s at the right level, ie. application level packet inspection; (b) it states “packet inspection” and fulfils the conditions:
-“LEAST operational overhead”
-“…to inspect all traffic to the application before the traffic reaches the web”

GLB won’t do it – because it states “receive the incoming packets and forward the packets to the appliance” – ie. NO inspection: the application gets the packet (good or bad!).

upvoted 1 times

Offset 9 months, 3 weeks ago

I agree with you as long as the question didn't mention anything about scalability and high availability for the network appliance.

upvoted 1 times

bishtr3 11 months ago

D : Gateway Load balancer : use when you have virtual appliances like IDP/IPS(instruction detection, prevention system..) & Firewall etc

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #20 *Topic 1*

A company wants to improve its ability to clone large amounts of production data into a test environment in the same AWS Region. The data is stored in Amazon EC2 instances on Amazon Elastic Block Store (Amazon EBS) volumes. Modifications to the cloned data must not affect the production environment. The software that accesses this data requires consistently high I/O performance.
A solutions architect needs to minimize the time that is required to clone the production data into the test environment.
Which solution will meet these requirements?

A. Take EBS snapshots of the production EBS volumes. Restore the snapshots onto EC2 instance store volumes in the test environment.

B. Configure the production EBS volumes to use the EBS Multi-Attach feature. Take EBS snapshots of the production EBS volumes. Attach the production EBS volumes to the EC2 instances in the test environment.

C. Take EBS snapshots of the production EBS volumes. Create and initialize new EBS volumes. Attach the new EBS volumes to EC2 instances in the test environment before restoring the volumes from the production EBS snapshots.

D. Take EBS snapshots of the production EBS volumes. Turn on the EBS fast snapshot restore feature on the EBS snapshots. Restore the snapshots into new EBS volumes. Attach the new EBS volumes to EC2 instances in the test environment. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (93%) Other (7%)

Comments

UWSFish Highly Voted 2 years, 7 months ago

Selected Answer: D

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-fast-snapshot-restore.html

Amazon EBS fast snapshot restore (FSR) enables you to create a volume from a snapshot that is fully initialized at creation. This eliminates the latency of I/O operations on a block when it is accessed for the first time. Volumes that are created using fast snapshot restore instantly deliver all of their provisioned performance.

upvoted 54 times

PhucVuu Highly Voted 8 months, 3 weeks ago

Selected Answer: D

Selected Answer: D

Keywords:
- Modifications to the cloned data must not affect the production environment.
- Minimize the time that is required to clone the production data into the test environment.

A: Incorrect - we can do this But it is not minimize the time as requirement.
B: Incorrect - This approach use same EBS volumes for produciton and test. If we modify test then it will be affected prodution environment.
C: Incorrect - EBS snapshot will create new EBS volumes. It can not restore from existing volumes.
D: Correct - Turn on the EBS fast snapshot restore feature on the EBS snapshots -> no latency on first use

upvoted 43 times

se7en102 Most Recent 1 month, 3 weeks ago

Selected Answer: D

EBS Snapshots: Taking snapshots of the production EBS volumes allows you to create point-in-time backups of the data without impacting the production environment.
Fast Snapshot Restore: Enabling the EBS fast snapshot restore feature allows you to restore the snapshots to new EBS volumes much more quickly than standard restoration. This significantly reduces the time required to clone the data.
New EBS Volumes: By restoring the snapshots into new EBS volumes, you ensure that the test environment is completely isolated from the production environment, allowing for modifications without any risk to production data.
High I/O Performance: EBS volumes provide high I/O performance, which is essential for the software accessing the data in the test environment.

upvoted 1 times

PaulGa 10 months ago

Selected Answer: D

Ans D - as per PhucVuu response... what's to debate...

upvoted 1 times

1e22522 10 months, 2 weeks ago

Selected Answer: D

Ye its d cuh

upvoted 1 times

1dfed2b 1 year, 3 months ago

Selected Answer: C

https://docs.aws.amazon.com/ebs/latest/userguide/ebs-restoring-volume.html
Its C. reate a new volume from the snapshot. Use the create-volume command. For --snapshot-id, specify the ID of the snapshot to use. For --availability-zone, specify the same Availability Zone as the instance. Configure the remaining parameters as needed.

upvoted 2 times

Isomas 1 year, 3 months ago

Answer is D because volumes that are created using fast snapshot restore instantly deliver all of their provisioned performance. Volumes created from normal snapshots will take time to initialize

upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: D

A: Can work but long cloning time
B: Wrong as multi attach will mean changes by test will affect production
C: Slow
D: Fast restore makes this a quicker option

upvoted 3 times

A_jaa 1 year, 5 months ago

Selected Answer: D

Answer-D

upvoted 1 times

Ruffyit 1 year, 7 months ago

Amazon EBS fast snapshot restore (FSR) enables you to create a volume from a snapshot that is fully initialized at creation. This

Amazon EBS fast snapshot restore (FSR) enables you to create a volume from a snapshot that is fully initialized at creation. This eliminates the latency of I/O operations on a block when it is accessed for the first time. Volumes that are created using fast snapshot restore instantly deliver all of their provisioned performance.

upvoted 1 times

ukivanlamlpi 1 year, 10 months ago

Selected Answer: A

why not A? high I/O, no need durability

upvoted 2 times

JackLo 1 year, 9 months ago

Although it is test environment, it's data should be durable

upvoted 3 times

TariqKipkemei 1 year, 10 months ago

Selected Answer: D

Needs to minimize the time that is required to clone the production data into the test environment = EBS fast snapshot restore feature

upvoted 1 times

Anil_Awasthi 1 year, 10 months ago

Selected Answer: C

Option C provides an effective solution for cloning large amounts of production data into a test environment with minimized time, high I/O performance, and without affecting the production environment.

upvoted 1 times

pentium75 1 year, 5 months ago

But you don't need a new, empty volume, you need a restore of the PROD snapshot. Thus D.

upvoted 2 times

Guru4Cloud 1 year, 10 months ago

Selected Answer: D

The correct answer is D.
Here is a step-by-step explanation of how to clone production data into a test environment using EBS snapshots:
Take EBS snapshots of the production EBS volumes.
Turn on the EBS fast snapshot restore feature on the EBS snapshots.
Restore the snapshots into new EBS volumes.
Attach the new EBS volumes to EC2 instances in the test environment.
The EBS fast snapshot restore feature allows you to restore snapshots more quickly than the default method. This is because the feature uses a process called parallel restore, which allows multiple EBS volumes to be restored at the same time.
The EBS fast snapshot restore feature is only available for EBS snapshots that are created in the same AWS Region as the EC2 instances that you are using to restore the snapshots.

upvoted 6 times

Thornessen 1 year, 11 months ago

For consistently high IO, option A is the solution. Instance store has the highest IO

upvoted 1 times

idanr391 1 year, 11 months ago

Its not, D its the solution. https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-fast-snapshot-restore.html

upvoted 1 times

miki111 1 year, 11 months ago

Option D is the ideal answer.

upvoted 1 times

cookieMr 1 year, 12 months ago

Selected Answer: D

Take EBS snapshots of the production EBS volumes. Turn on the EBS fast snapshot restore feature on the EBS snapshots

Take EBS snapshots of the production EBS volumes. Turn on the EBS fast snapshot restore feature on the EBS snapshots. Restore the snapshots into new EBS volumes. Attach the new EBS volumes to EC2 instances in the test environment.

Enabling the EBS fast snapshot restore feature allows you to restore EBS snapshots into new EBS volumes almost instantly, without needing to wait for the data to be fully copied from the snapshot. This significantly reduces the time required to clone the production data.

By taking EBS snapshots of the production EBS volumes and restoring them into new EBS volumes in the test environment, you can ensure that the cloned data is separate and does not affect the production environment. Attaching the new EBS volumes to the EC2 instances in the test environment allows you to access the cloned data.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #21 *Topic 1*

An ecommerce company wants to launch a one-deal-a-day website on AWS. Each day will feature exactly one product on sale for a period of 24 hours. The company wants to be able to handle millions of requests each hour with millisecond latency during peak hours.
Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon S3 to host the full website in different S3 buckets. Add Amazon CloudFront distributions. Set the S3 buckets as origins for the distributions. Store the order data in Amazon S3.

B. Deploy the full website on Amazon EC2 instances that run in Auto Scaling groups across multiple Availability Zones. Add an Application Load Balancer (ALB) to distribute the website traffic. Add another ALB for the backend APIs. Store the data in Amazon RDS for MySQL.

C. Migrate the full application to run in containers. Host the containers on Amazon Elastic Kubernetes Service (Amazon EKS). Use the Kubernetes Cluster Autoscaler to increase and decrease the number of pods to process bursts in traffic. Store the data in Amazon RDS for MySQL.

D. Use an Amazon S3 bucket to host the website's static content. Deploy an Amazon CloudFront distribution. Set the S3 bucket as the origin. Use Amazon API Gateway and AWS Lambda functions for the backend APIs. Store the data in Amazon DynamoDB. **Most Voted**

Correct Answer: *D*

Community vote distribution

D (100%)

Comments

Sinaneos Highly Voted 2 years, 8 months ago

Selected Answer: D

D because all of the components are infinitely scalable
dynamoDB, API Gateway, Lambda, and of course s3+cloudfront

upvoted 47 times

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: D

The solution that will meet these requirements with the least operational overhead is D: Use an Amazon S3 bucket to host the website's static content, deploy an Amazon CloudFront distribution, set the S3 bucket as the origin, and use Amazon API Gateway and AWS Lambda functions for the backend APIs. Store the data in Amazon DynamoDB.

Using Amazon S3 to host static content and Amazon CloudFront to distribute the content can provide high performance and scale for websites with millions of requests each hour. Amazon API Gateway and AWS Lambda can be used to build scalable and highly available backend APIs to support the website, and Amazon DynamoDB can be used to store the data. This solution requires minimal operational overhead as it leverages fully managed services that automatically scale to meet demand.

upvoted 22 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option A is incorrect because using multiple S3 buckets to host the full website would not provide the required performance and scale for millions of requests each hour with millisecond latency.

Option B is incorrect because deploying the full website on EC2 instances and using an Application Load Balancer (ALB) and an RDS database would require more operational overhead to maintain and scale the infrastructure.

Option C is incorrect because while deploying the application in containers and hosting them on Amazon Elastic Kubernetes Service (EKS) can provide high performance and scale, it would require more operational overhead to maintain and scale the infrastructure compared to using fully managed services like S3 and CloudFront.

upvoted 23 times

try1260 6 months, 3 weeks ago

How to use dynmodb in a emcommerec trx it requrted sql databse

upvoted 2 times

phenic Most Recent 1 month, 1 week ago

Selected Answer: D

The correct answer is D . You use s3 to host static contect. the cloudfront will CloudFront to distribute the content can provide high performance and scale for websites with millions of requests each hour.Amazon API Gateway and AWS Lambda can be used to build scalable and highly available backend APIs to support the website, and Amazon DynamoDB can be used to store the data. Amazon API Gateway and AWS Lambda can be used to build scalable and highly available backend APIs to support the website, and Amazon DynamoDB can be used to store the data.

upvoted 1 times

AshishDhole 5 months ago

Selected Answer: D

Millisecond Latency = DynamoDB

upvoted 3 times

satyaammm 5 months, 2 weeks ago

Selected Answer: D

D as DynamoDB is only suitable because of its really fast milliseconds speed.

upvoted 1 times

EzKkk 7 months ago

Selected Answer: D

Requirement analysis:
- Build an e commerce application -> Typical 3 tiers app.
- Deployment.
- Scale quickly.
- Least operational overhead.

Based on what we get from the question, option D is quite self explanatory.
S3 - Static resources storage, this can also used to deploy static web page if SPA is not desired
CloudFront - Distribute application through edge locations
API Gateway, Lambda, and DynamoDB - Simple application and db layers, scale quickly, serverless so no operational burden

upvoted 2 times

try1260 6 months, 3 weeks ago

How to use dynmodb in a emcommerec trx it requrted sql databse

upvoted 2 times

PaulEkwem 8 months ago

Option D:
Use Amazon S3 to store the website's static content (like images, HTML, etc.). Static content doesn't change based on user input, so S3 is perfect because it's highly scalable and can handle millions of requests.
Use Amazon CloudFront to distribute the content globally, reducing latency by delivering it from servers close to the user.
For backend operations (like placing orders), use API Gateway and Lambda functions. API Gateway handles incoming requests, while Lambda runs code without needing servers (it's serverless), which automatically scales to meet the traffic demand.
Store the data in DynamoDB: DynamoDB is a fully managed, NoSQL database that can handle huge traffic spikes with very fast performance, which is important for millions of visitors.

upvoted 3 times

try1260 6 months, 3 weeks ago

How to use dynmodb in a emcommerec trx it requrted sql databse

upvoted 1 times

qmailmadrid 9 months, 2 weeks ago

My first answer is C. Coz it's high performance and high scalability.
I didn't get the D answer:
(1) So S3 can host static website? also is it mentioned on the question?
(2) Lambda, S3, DynamoDB, all service managed by AWS, but so what, same with (1), which component host the main page?

upvoted 1 times

bishtr3 11 months ago

D operational overhead
Lamda is a serverless computing let you run code without provisioning or managing service
API Gateway -> Lamda -> Dynamo DB

upvoted 2 times

otaku2398 11 months, 1 week ago

can someone pls tell me how d is the answer. doesn't lambda time out in 15min

upvoted 3 times

ramkinkarpandey 1 year ago

While everyone is voting for D but no where in the question it mentions that website is made of static pages. Other than not mentioning static, option D checks all the boxes.

upvoted 4 times

JohnZh 1 year, 2 months ago

I struggled between A and D a little bit, then realize that A is not correct because it's hosting website in "different" buckets.

upvoted 4 times

awsgeek75 1 year, 5 months ago

Selected Answer: D

Least operational overhead is only possible with managed services that deliver the required solution.
A: Cannot store order data in S3 as there is no processing in S3
B: Overhead of EC2 and RDS and ALB, too many moving parts
C: Container management is overhead and RDS too
D: S3 for static is best practice. CloudFront helps with scaling. API GW with Lambda is fully managed. DynamoDB for transactions is managed scalable solution.

upvoted 3 times

A_jaa 1 year, 5 months ago

Selected Answer: D

Answer-D

upvoted 1 times

bujuman 1 year, 5 months ago

Selected Answer: D

D is the best asnwer for least operation

upvoted 1 times

upvoted 1 times

ddement0r 1 year, 6 months ago

Selected Answer: D

D because it is the most logical solution

upvoted 1 times

Avirexirex 1 year, 6 months ago

Selected Answer: D

correct answer

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #22 *Topic 1*

A solutions architect is using Amazon S3 to design the storage architecture of a new digital media application. The media files must be resilient to the loss of an Availability Zone. Some files are accessed frequently while other files are rarely accessed in an unpredictable pattern. The solutions architect must minimize the costs of storing and retrieving the media files.
Which storage option meets these requirements?

A. S3 Standard
B. S3 Intelligent-Tiering **Most Voted**
C. S3 Standard-Infrequent Access (S3 Standard-IA)
D. S3 One Zone-Infrequent Access (S3 One Zone-IA)

Correct Answer: *B*

Community vote distribution

B (100%)

Comments

123jhl0 Highly Voted 2 years, 8 months ago

Selected Answer: B

"unpredictable pattern" - always go for Intelligent Tiering of S3
It also meets the resiliency requirement: "S3 Standard, S3 Intelligent-Tiering, S3 Standard-IA, S3 Glacier Instant Retrieval, S3 Glacier Flexible Retrieval, and S3 Glacier Deep Archive redundantly store objects on multiple devices across a minimum of three Availability Zones in an AWS Region" https://docs.aws.amazon.com/AmazonS3/latest/userguide/DataDurability.html
upvoted 49 times

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: B

The storage option that meets these requirements is B: S3 Intelligent-Tiering.

Amazon S3 Intelligent Tiering is a storage class that automatically moves data to the most cost-effective storage tier based on access patterns. It can store objects in two access tiers: the frequent access tier and the infrequent access tier. The frequent access tier is optimized for frequently accessed objects and is charged at the same rate as S3 Standard. The infrequent access tier is optimized for objects that are not accessed frequently and are charged at a lower rate than S3 Standard.

S3 Intelligent Tiering is a good choice for storing media files that are accessed frequently and infrequently in an unpredictable pattern because it automatically moves data to the most cost-effective storage tier based on access patterns, minimizing storage and retrieval costs. It is also resilient to the loss of an Availability Zone because it stores objects in multiple Availability Zones within a region.

upvoted 19 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option A, S3 Standard, is not a good choice because it does not offer the cost optimization of S3 Intelligent-Tiering.

Option C, S3 Standard-Infrequent Access (S3 Standard-IA), is not a good choice because it is optimized for infrequently accessed objects and does not offer the cost optimization of S3 Intelligent-Tiering.

Option D, S3 One Zone-Infrequent Access (S3 One Zone-IA), is not a good choice because it is not resilient to the loss of an Availability Zone. It stores objects in a single Availability Zone, making it less durable than other storage classes.

upvoted 8 times

LEEOscar Most Recent 4 weeks, 1 day ago

Selected Answer: B

Amazon S3 Intelligent-Tiering:

Automatically moves data between different storage tiers based on access patterns.

Designed for unpredictable access—you don't need to know in advance whether data will be frequently accessed.

Provides multi-AZ durability, just like S3 Standard.

upvoted 1 times

San757 1 month ago

Selected Answer: B

The storage option that meets these requirements is B: S3 Intelligent-Tiering.

upvoted 1 times

Gizmo2022 6 months ago

Selected Answer: B

I am going to choose B.
S3 Intelligent-Tiering - Perfect use case when you don't know the frequency of access or irregular patterns of usage.

upvoted 1 times

Tjazz04 6 months ago

Selected Answer: B

Key phrase: Some files are accessed frequently while other files are rarely accessed in an unpredictable pattern.
Always use S3 Intelligent-Tiering for unpredictable access patterns.

upvoted 1 times

PaulGa 10 months ago

Selected Answer: B

Ans B - Intelligent Tiering: cost effective, optimised by access frequency

upvoted 2 times

DavidNgTan 10 months, 3 weeks ago

Selected Answer: B

S3 intelligent tiering support 3 layer, frequent access, infrequent access and rarely access data.
https://docs.aws.amazon.com/AmazonS3/latest/userguide/intelligent-tiering-overview.html

upvoted 3 times

lofzee 1 year ago

Unpredictable pattern equals intelligent tiering

upvoted 3 times

JohnZh 1 year, 2 months ago

unpredictable pattern == S3 IA (Intellgent Tiering, not Infrequent access though)

upvoted 3 times

awsgeek75 1 year, 5 months ago

Selected Answer: B

Unpredictable pattern = Intelligent tiering

upvoted 3 times

A_jaa 1 year, 5 months ago

Selected Answer: B

Answer-B

upvoted 1 times

bujuman 1 year, 5 months ago

Selected Answer: B

The right answer due to "unpredictable pattern"

upvoted 1 times

ddement0r 1 year, 6 months ago

Selected Answer: B

B because intelligent tiering is what we choose when we don't have a pattern

upvoted 2 times

Ruffyit 1 year, 7 months ago

Amazon S3 Intelligent Tiering is a storage class that automatically moves data to the most cost-effective storage tier based on access patterns. It can store objects in two access tiers: the frequent access tier and the infrequent access tier. The frequent access tier is optimized for frequently accessed objects and is charged at the same rate as S3 Standard. The infrequent access tier is optimized for objects that are not accessed frequently and are charged at a lower rate than S3 Standard.

upvoted 2 times

awsleffe 1 year, 8 months ago

(B) The question mentions that some files are accessed frequently while others are rarely accessed, and the pattern is unpredictable.
This makes S3 Intelligent-Tiering a good fit because it automatically moves data between different access tiers based on how frequently they are accessed, optimizing costs.
Intelligent-Tiering is designed to optimize costs by automatically moving data to the most cost-effective access tier, without performance impact or operational overhead.
B meets the requirements

upvoted 1 times

reema908516 1 year, 9 months ago

Selected Answer: B

Amazon S3 Intelligent Tiering is a storage class that automatically moves data to the most cost-effective storage tier based on access patterns.

upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #23 *Topic 1*

A company is storing backup files by using Amazon S3 Standard storage. The files are accessed frequently for 1 month. However, the files are not accessed after 1 month. The company must keep the files indefinitely.
Which storage solution will meet these requirements MOST cost-effectively?

A. Configure S3 Intelligent-Tiering to automatically migrate objects.

B. Create an S3 Lifecycle configuration to transition objects from S3 Standard to S3 Glacier Deep Archive after 1 month.
Most Voted

C. Create an S3 Lifecycle configuration to transition objects from S3 Standard to S3 Standard-Infrequent Access (S3 Standard-IA) after 1 month.

D. Create an S3 Lifecycle configuration to transition objects from S3 Standard to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 1 month.

Correct Answer: *B*

Community vote distribution

B (96%) Other

Comments

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: B

The storage solution that will meet these requirements most cost-effectively is B: Create an S3 Lifecycle configuration to transition objects from S3 Standard to S3 Glacier Deep Archive after 1 month.

Amazon S3 Glacier Deep Archive is a secure, durable, and extremely low-cost Amazon S3 storage class for long-term retention of data that is rarely accessed and for which retrieval times of several hours are acceptable. It is the lowest-cost storage option in Amazon S3, making it a cost-effective choice for storing backup files that are not accessed after 1 month.

You can use an S3 Lifecycle configuration to automatically transition objects from S3 Standard to S3 Glacier Deep Archive after 1 month. This will minimize the storage costs for the backup files that are not accessed frequently.

upvoted 17 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option A, configuring S3 Intelligent-Tiering to automatically migrate objects, is not a good choice because it is not designed for long-term storage and does not offer the cost benefits of S3 Glacier Deep Archive.

Option C, transitioning objects from S3 Standard to S3 Standard-Infrequent Access (S3 Standard-IA) after 1 month, is not a good choice because it is not the lowest-cost storage option and would not provide the cost benefits of S3 Glacier Deep Archive.

Option D, transitioning objects from S3 Standard to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 1 month, is not a good choice because it is not the lowest-cost storage option and would not provide the cost benefits of S3 Glacier Deep Archive.

upvoted 7 times

JA2018 7 months, 1 week ago

If the AZ selected for the One Zone-IA option is whacked offline, high chances of data loss.

upvoted 1 times

vgchan 2 years, 5 months ago

Also S3 Standard-IA & One Zone-IA stores the data for max of 30 days and not indefinitely.

upvoted 6 times

MatAlves 8 months, 3 weeks ago

you want the MOST cost-effectively solution. Glacier DA is the cheapest for archival.

upvoted 1 times

ninjawrz Highly Voted 2 years, 8 months ago

B: Transition to Glacier deep archive for cost efficiency

upvoted 7 times

phenic Most Recent 1 month, 1 week ago

Selected Answer: B

Amazon S3 Glacier Deep Archive is a secure, durable, and extremely low-cost Amazon S3 storage class for long-term retention of data that is rarely accessed and for which retrieval times of several hours are acceptable. It is the lowest-cost storage option in Amazon S3, making it a cost-effective choice for storing backup files that are not accessed after 1 month.

upvoted 1 times

Vandaman 3 months, 3 weeks ago

Selected Answer: B

Definitely B. Files are not accessed after 1 month and need to be kept indefinitely, so Glacier Deep Archive is the best solution.

upvoted 1 times

satyaammm 5 months, 2 weeks ago

Selected Answer: B

Glacier archive is most suitable here as the data doesn't needs to be accessed anymore but stored indefinitely.

upvoted 1 times

LayManCloud_2050 7 months, 1 week ago

I think the answer is C. This is because, the files were accessed frequently for the first month, so transitioning to Glacier Deep Archive after 1 month would result in higher retrieval costs for the first month of frequent access, making it a less optimal solution. However, since the files are accessed frequently for the first month and then not accessed after that, transitioning them to S3 Standard-IA after 1 month is the cost-effective choice.

upvoted 1 times

PaulGa 10 months ago

Selected Answer: B

Ans B - Glacier: the files will not be accessed after 1 month; they just need to be retained

upvoted 1 times

DavidNgTan 10 months, 3 weeks ago

Selected Answer: B

Should apply S3 lifecycle to move not accessed file after 1 month to S3 Glacier Deep Archive.

upvoted 1 times

Solomon2001 1 year, 1 month ago

Selected Answer: C

Since the files are not accessed after 1 month but need to be kept indefinitely, transitioning them to S3 Standard-Infrequent Access (S3 Standard-IA) would be the best choice.

upvoted 1 times

JohnZh 1 year, 2 months ago

not accessed == Galcier -- easy one

upvoted 1 times

awsgeek75 1 year, 5 months ago

Selected Answer: B

A: Possible but expensive
CD: One zone so no guarantee of being stored indefinitely.
B: S3GDA is cost effective indefinite storage

upvoted 1 times

A_jaa 1 year, 5 months ago

Selected Answer: B

Answer-B

upvoted 1 times

RNess 1 year, 5 months ago

It's can't be B!!
because objects that are archived to S3 Glacier Instant Retrieval and S3 Glacier Flexible Retrieval are charged for a minimum storage duration of 90 days, and S3 Glacier Deep Archive has a minimum storage duration of 180 days.

upvoted 2 times

RichWil 3 months, 4 weeks ago

In AWS S3 Standard, there is no minimum storage duration requirement before you can move an object to S3 Glacier Deep Archive. You can transition objects at any time using S3 Lifecycle Policies.

However, once an object is stored in S3 Glacier Deep Archive, it has a minimum storage duration of 180 days. If the object is deleted or overwritten before 180 days, you will be charged for the full 180-day period.

upvoted 1 times

pentium75 1 year, 5 months ago

But the items must be kept forever, so where's the issue with that?

upvoted 1 times

RNess 1 year, 5 months ago

I mean, that min duration **before** can move to S3 Glacier Flexible Retrieval or S3 Glacier Deep Archive

upvoted 2 times

ddement0r 1 year, 6 months ago

Selected Answer: B

B because since the files should be kept but never accessed we can put them in Deep Archive

upvoted 1 times

Ruffyit 1 year, 7 months ago

Amazon S3 Glacier Deep Archive is a secure, durable, and extremely low-cost Amazon S3 storage class for long-term retention of data that is rarely accessed and for which retrieval times of several hours are acceptable

upvoted 1 times

AhmedAbdelhedi 1 year, 8 months ago

Selected Answer: B

Answer is B

upvoted 1 times

sujanakakarla 1 year, 9 months ago

Selected Answer: B

B as these files will be stored indefinitely after 1 month

upvoted 1 times

EXAMTOPICS

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #24 *Topic 1*

A company observes an increase in Amazon EC2 costs in its most recent bill. The billing team notices unwanted vertical scaling of instance types for a couple of EC2 instances. A solutions architect needs to create a graph comparing the last 2 months of EC2 costs and perform an in-depth analysis to identify the root cause of the vertical scaling.
How should the solutions architect generate the information with the LEAST operational overhead?

A. Use AWS Budgets to create a budget report and compare EC2 costs based on instance types.

B. Use Cost Explorer's granular filtering feature to perform an in-depth analysis of EC2 costs based on instance types.
Most Voted

C. Use graphs from the AWS Billing and Cost Management dashboard to compare EC2 costs based on instance types for the last 2 months.

D. Use AWS Cost and Usage Reports to create a report and send it to an Amazon S3 bucket. Use Amazon QuickSight with Amazon S3 as a source to generate an interactive graph based on instance types.

Correct Answer: *B*

Community vote distribution

B (65%) C (25%) D (10%)

Comments

sba21 Highly Voted 2 years, 8 months ago
Selected Answer: B

https://www.examtopics.com/discussions/amazon/view/68306-exam-aws-certified-solutions-architect-associate-saa-c02/
upvoted 38 times

123jhl0 Highly Voted 2 years, 8 months ago
Selected Answer: C

The requested result is a graph, so...
A - can't be as the result is a report
B - can't be as it is limited to 14 days visibility and the graph has to cover 2 months
C - seems to provide graphs and the best option available, as...
D - could provide graphs, BUT involves operational overhead, which has been requested to be minimised.
upvoted 31 times

satyaammm 5 months, 2 weeks ago

Hey brother, I think you might be wrong. The AWS Cost Explorer retains for 12 months and the 14 days visibility is actually for the free trial. So it has nothing related to how much the cost explorer can retain. I hope I made myself clear.
upvoted 3 times

kidomaruto 1 year, 7 months ago

"The Cost Explorer Hourly and Resource level granularity allows you to access cost and usage data at hourly granularity for the past 14 days and resource level granularity."

https://aws.amazon.com/fr/aws-cost-management/aws-cost-explorer/pricing/#:~:text=The%20Cost%20Explorer%20Hourly%20and,available%20for%20EC2%20instances%20only.
upvoted 2 times

Udoyen 2 years, 6 months ago

Cost Explorer, AWS prepares the data about your costs for the current month and the last 12 months:
https://aws.amazon.com/aws-cost-management/aws-cost-explorer/
upvoted 17 times

JA2018 7 months, 1 week ago

https://docs.aws.amazon.com/cost-management/latest/userguide/differences-billing-data-cost-explorer-data.html

Cost Explorer supports deep-dive analysis so that you can identify savings opportunities. Cost Explorer data provides more granular dimensions (such as Availability Zone or operating system) and includes features that might show differences when compared to billing data. On the Cost Management preferences page, you can manage your preferences for Cost Explorer data, including linked account access and historical and granular data settings. For more information, see Controlling access to Cost Explorer.
upvoted 1 times

goku58 2 years, 7 months ago

12 months data visible on Cost Explorer.
upvoted 17 times

Gongs Most Recent 1 month, 2 weeks ago

Selected Answer: C

C cause Timescale in b is wrong
upvoted 1 times

Juju66 1 month, 3 weeks ago

Selected Answer: B

C can't provide granular filter.
upvoted 1 times

Vandaman 3 months, 3 weeks ago

Selected Answer: B

I see lately that you can look back as far as 3 years.
upvoted 1 times

AshishDhole 4 months, 1 week ago

Selected Answer: B

Cost Explorer, AWS prepares the data about your costs for the current month and the last 12 months, and then calculates the forecast for the next 12 months. The current month's data is available for viewing in about 24 hours. The rest of your data takes a few days longer. Cost Explorer updates your cost data at least once every 24 hours.
upvoted 1 times

V2910 4 months, 3 weeks ago

Selected Answer: B

Cost explorer has option to filter by service and we can modify the timeline also to compare the costs across 3 months
upvoted 1 times

aatikah 6 months ago

Selected Answer: B

AWS Cost Explorer allows you to view cost and usage data with granular filtering. However, the historical data retention depends on the level of granularity:

Daily granularity: You can access up to the past 12 months of daily cost and usage data.

Hourly granularity: If you have detailed billing enabled, you can view up to the past 14 days of hourly usage data.

Monthly granularity: You can access up to the past 12 months of cost and usage data, and also forecast costs for the next 12 months.

upvoted 3 times

mzeynalli 7 months, 3 weeks ago

Selected Answer: B

B. Use Cost Explorer's granular filtering feature to perform an in-depth analysis of EC2 costs based on instance types.

Explanation:
Cost Explorer provides an easy way to analyze AWS costs and usage visually. It allows you to filter data by multiple parameters, such as instance types, regions, and time periods. It also enables you to drill down and identify specific cost drivers, such as unwanted vertical scaling.
This option is suitable because it provides granular insights with minimal operational overhead, as it is a built-in AWS tool specifically designed for cost analysis.

upvoted 4 times

mzeynalli 7 months, 3 weeks ago

The correct answer is:

B. Use Cost Explorer's granular filtering feature to perform an in-depth analysis of EC2 costs based on instance types.

Explanation:

Cost Explorer provides an easy way to analyze AWS costs and usage visually. It allows you to filter data by multiple parameters, such as instance types, regions, and time periods. It also enables you to drill down and identify specific cost drivers, such as unwanted vertical scaling.
This option is suitable because it provides granular insights with minimal operational overhead, as it is a built-in AWS tool specifically designed for cost analysis.
Why Other Options Are Less Suitable:

C. AWS Billing and Cost Management Dashboard: It provides a high-level view of costs, but it lacks the specific filtering capabilities needed for a granular, instance-type-level analysis over time.

upvoted 2 times

PaulEkwem 8 months ago

Option B:
Use AWS Cost Explorer, a tool that makes it easy to view your costs and break them down into categories like EC2 instance types, time periods, and more.
With Cost Explorer, you can easily filter the information to see which EC2 instance types caused the increase in costs over the last two months. This lets you do an in-depth analysis without much effort.

upvoted 3 times

PaulGa 10 months ago

Selected Answer: B

I would have gone for Ans B but apparently the right one is Ans C.
I'm not convinced because neither B or C actually determine the root cause – they just point you in the right direction and then you'll need to do some further analysis around resource demand (CPU, storage, network, etc), data/network traffic, what function/ instructions are actually being processed, along with taking a view of the scaling algorithms. On that basis I'd have said Ans B because it requires the LEAST overhead to get to the next step which is the one that matters: the root analysis for vertical scaling.

upvoted 3 times

DavidNgTan 10 months, 3 weeks ago

Selected Answer: B

AWS Cost explorer will provide your usage and cost by main graph.
https://docs.aws.amazon.com/cost-management/latest/userguide/ce-what-is.html

upvoted 2 times

upvoted 2 times

stlwell 1 year ago

Selected Answer: C

I think the highest priority is "with the LEAST operational overhead?". B is very good for "perform an in-depth analysis" but C overwhelming win on cost.

upvoted 2 times

firsttimetesttaker 1 year, 2 months ago

Selected Answer: D

Both B and D has their merits and achieve the ask of a question. Infact Option D would give more streamlined and automated approach and will be very less overhead once setup.

upvoted 1 times

JohnZh 1 year, 2 months ago

Not sure why it's B -- how can cost explorer identify the root cause of the vertical scaling?

upvoted 1 times

JohnZh 1 year, 2 months ago

Oh I see why: they want to identify "unwanted vertical scaling of instance types for a couple of EC2 instances", which could be RDS, ES, ElasticCache, and etc:
https://aws.amazon.com/aws-cost-management/aws-cost-explorer/features/#:~:text=Cost%20Explorer%20allows%20customers%20to,or%20understand%20peak%20hour%20usage

upvoted 2 times

ManikRoy 1 year, 2 months ago

Selected Answer: B

Note that If Hourly granularity is required then the correct option might not be B as cost explorer hourly granular details are provided only for past 14 days.
reference link -
https://aws.amazon.com/aws-cost-management/aws-cost-explorer/features/#:~:text=Cost%20Explorer%20allows%20customers%20to,or%20understand%20peak%20hour%20usage.

upvoted 2 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #25 *Topic 1*

A company is designing an application. The application uses an AWS Lambda function to receive information through Amazon API Gateway and to store the information in an Amazon Aurora PostgreSQL database.
During the proof-of-concept stage, the company has to increase the Lambda quotas significantly to handle the high volumes of data that the company needs to load into the database. A solutions architect must recommend a new design to improve scalability and minimize the configuration effort.
Which solution will meet these requirements?

A. Refactor the Lambda function code to Apache Tomcat code that runs on Amazon EC2 instances. Connect the database by using native Java Database Connectivity (JDBC) drivers.

B. Change the platform from Aurora to Amazon DynamoDProvision a DynamoDB Accelerator (DAX) cluster. Use the DAX client SDK to point the existing DynamoDB API calls at the DAX cluster.

C. Set up two Lambda functions. Configure one function to receive the information. Configure the other function to load the information into the database. Integrate the Lambda functions by using Amazon Simple Notification Service (Amazon SNS).

D. Set up two Lambda functions. Configure one function to receive the information. Configure the other function to load the information into the database. Integrate the Lambda functions by using an Amazon Simple Queue Service (Amazon SQS) queue. Most Voted

Correct Answer: *D*

Community vote distribution

D (99%) B

Comments

123jhl0 Highly Voted 2 years, 8 months ago

Selected Answer: D

A - refactoring can be a solution, BUT requires a LOT of effort - not the answer
B - DynamoDB is NoSQL and Aurora is SQL, so it requires a DB migration... again a LOT of effort, so no the answer
C and D are similar in structure, but...
C uses SNS, which would notify the 2nd Lambda function... provoking the same bottleneck... not the solution
D uses SQS, so the 2nd lambda function can go to the queue when responsive to keep with the DB load process.
Usually the app decoupling helps with the performance improvement by distributing load. In this case, the bottleneck is solved by uses queues... so D is the answer.

upvoted 103 times

PhucVuu Highly Voted 2 years, 2 months ago

Selected Answer: D

Keywords:
- Company has to increase the Lambda quotas significantly to handle the high volumes of data that the company needs to load into the database.
- Improve scalability and minimize the configuration effort.

A: Incorrect - Lambda is Serverless and automatically scale - EC2 instance we have to create load balancer, auto scaling group,.. a lot of things. using native Java Database Connectivity (JDBC) drivers don't improve the performance.
B: Incorrect - a lot of things to changes and DynamoDB Accelerator use for cache(read) not for write.
C: Incorrect - SNS is use for send notification (e-mail, SMS).
D: Correct - with SQS we can scale application well by queuing the data.

upvoted 21 times

PaulGa Most Recent 10 months ago

Selected Answer: D

It could be C or D, but Ans D wins because
–SNS is Push Mechanism - 'Other Lambda' function is forced to take message when it might not be ready (or refuse it)
–SQS is Pull Mechanism – 'Other Lambda' function can take next message when its ready to do so
SQS is simple and allows better de-coupling.

upvoted 2 times

rodrigoleoncio 1 year, 1 month ago

Selected Answer: D

uses SQS

upvoted 1 times

JohnZh 1 year, 2 months ago

If the throughput is so high that lambda concurrency needs to go beyond 1000, we need to set up a queue to throttle the request.

upvoted 3 times

TheFivePips 1 year, 4 months ago

Selected Answer: D

D. Set up two Lambda functions, one for receiving information and another for loading data into the database. Integrate them using an Amazon SQS queue. This approach allows for better scalability, maintains the serverless paradigm, and minimizes manual configuration effort. It leverages Amazon SQS as a reliable message queue between Lambda functions.

Options A and B introduce complexities and changes in architecture, while Option C introduces an additional service that may not be as suitable for decoupling processes in this scenario.

upvoted 2 times

awsgeek75 1 year, 5 months ago

Selected Answer: D

SQS will help Lambda scale even more.
A EC2 + Tomcat will be slower than Lambda for this usecase
B is wrong because the problem is with Lambda scaling not the DB
C SNS is not the best option for this usecase when SQS is an option

upvoted 5 times

A_jaa 1 year, 5 months ago

Selected Answer: D

Answer-D

upvoted 1 times

ddement0r 1 year, 6 months ago

Selected Answer: D

D : other ones just don't make sense

upvoted 1 times

pedestrianlove 1 year, 6 months ago

Sorry, but the question does not make sense by itself. What are you asking for more scalability from an already scalable Lambda function?

If you're concerned about the concurrency limits of Lambda function, decoupling just doesn't make sense, since it'll keep even more lambda instances running in a given time period(including 2 phases of execution for each request, let alone the cold start issues).

If you're concerned about bottleneck database induced, that'll even be more ridiculous since you're supposed to resolve the scalability issue of the database(e.g. Aurora) instead of decoupling the Lambda function to improve the throughput of this entire data flow.

upvoted 2 times

mohamedsambo 1 year, 5 months ago

i think it is clear that he want to enhance the lambda even more than "The default concurrency limit across all functions per region in a given account is 1,000"
cause sqs can scale and store the data till new available revoked lambda consume it

upvoted 3 times

Ruffyit 1 year, 7 months ago

Lambda and SQS are serverless. No involvement will be required in execution.

upvoted 2 times

xdkonorek2 1 year, 7 months ago

Selected Answer: B

I think B would be better solution.
How splitting one function into 2 increase scalability when company already increased service quota? Effectively they will have same compute time.
Changing Aurora to DAX will shorten the time for data loads by ~100x requiring way less time for data loading, and it's most time consuming thing this lambda does. DAX has better scaling than aurora and is better fit with lambda

upvoted 2 times

MakaylaLearns 1 year, 9 months ago

Lambda Functions: A review
Run your code in response to events

You can build chatbots using Lambda functions to process user input, execute business logic, and generate responses.
Scales automatically
They can be triggered in response to API events
Lambda functions can process files as they are uploaded to S3 buckets. This is often used for tasks like image resizing, data extraction, or file validation.

upvoted 1 times

[Removed] 1 year, 9 months ago

AWS Cost Explorer is a tool that enables you to view and analyze your costs and usage. You can explore your usage and costs using the main graph, the Cost Explorer cost and usage reports, or the Cost Explorer RI reports. You can view data for up to the last 12 months, forecast how much you're likely to spend for the next 12 months, and get recommendations for what Reserved Instances to purchase.
Ans: B is correct
https://docs.aws.amazon.com/cost-management/latest/userguide/ce-what-is.html

upvoted 1 times

doujones 1 year, 10 months ago

Do you all have to take the whole practice exam on here, in order to pass AWS SAA C03

upvoted 2 times

TariqKipkemei 1 year, 10 months ago

Increase Lambda quotas = Set up two Lambda functions. Improve scalability = Amazon Simple Queue Service.

upvoted 1 times

TariqKipkemei 1 year, 10 months ago

Selected answer D

upvoted 1 times

upvoted 1 times

miki111 1 year, 11 months ago

Option D is the right answer for this.
upvoted 1 times

Exam AWS Certified Solutions Architect - Associate SAA-C03 All Actual Questions

Question #26 *Topic 1*

A company needs to review its AWS Cloud deployment to ensure that its Amazon S3 buckets do not have unauthorized configuration changes.
What should a solutions architect do to accomplish this goal?

A. Turn on AWS Config with the appropriate rules. **Most Voted**

B. Turn on AWS Trusted Advisor with the appropriate checks.

C. Turn on Amazon Inspector with the appropriate assessment template.

D. Turn on Amazon S3 server access logging. Configure Amazon EventBridge (Amazon Cloud Watch Events).

Correct Answer: *A*

Community vote distribution

A (98%) B

Comments

Buruguduystunstugudunstuy Highly Voted 2 years, 5 months ago

Selected Answer: A

The solution that will accomplish this goal is A: Turn on AWS Config with the appropriate rules.

AWS Config is a service that enables you to assess, audit, and evaluate the configurations of your AWS resources. You can use AWS Config to monitor and record changes to the configuration of your Amazon S3 buckets. By turning on AWS Config and enabling the appropriate rules, you can ensure that your S3 buckets do not have unauthorized configuration changes.

upvoted 62 times

Buruguduystunstugudunstuy 2 years, 5 months ago

AWS Trusted Advisor (Option B) is a service that provides best practice recommendations for your AWS resources, but it does not monitor or record changes to the configuration of your S3 buckets.

Amazon Inspector (Option C) is a service that helps you assess the security and compliance of your applications. While it can be used to assess the security of your S3 buckets, it does not monitor or record changes to the configuration of your S3 buckets.

Amazon S3 server access logging (Option D) enables you to log requests made to your S3 bucket. While it can help you identify changes to your S3 bucket, it does not monitor or record changes to the configuration of your S3 bucket.

upvoted 49 times

gokalpkocer3 Highly Voted 2 years, 7 months ago